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An integrated explainable framework for multimodal deepfake detection across image, audio, and video data

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ABSTRACT

The rapid advancement of deepfake generation techniques has introduced significant challenges for digital forensics, particularly due to the increasing realism and multimodal nature of manipulated content. Existing detection approaches are predominantly unimodal and lack interpretability, limiting their effectiveness in real-world scenarios. To address these limitations, this study proposes a unified and explainable deepfake detection framework that operates across image, video, audio, and multimodal data. The proposed framework integrates modality-specific deep learning architectures, InceptionV3 for images, DenseNet169-Bidirectional Long Short-Term Memory (BiLSTM) for video, and Convolutional Neural Network (CNN)-BiLSTM with Extreme Gradient Boosting (XGBoost) for audio, along with a cross-attention-based fusion model for multimodal analysis. A key contribution of this work is a modality-adaptive explainable artificial intelligence (XAI) strategy that systematically combines Shapley Additive Explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), Layer-wise Relevance Propagation (LRP), and Gradient-weighted Class Activation Mapping (Grad-CAM) to generate consistent visual and textual explanations across different data types. Extensive experiments conducted on benchmark datasets, including 140 K-Faces, Celeb-DF-V2, In-The-Wild, and LAV-DF, demonstrate the effectiveness of the proposed approach, achieving accuracy rates of 99% for images, 96% for videos, 98% for audio, and 99% for multimodal data. Furthermore, the framework provides interpretable insights into model decisions through an interactive interface, enhancing transparency and usability in practical applications. The results indicate that the proposed system not only improves detection performance but also addresses the critical need for unified and interpretable deepfake analysis in multimodal environments.

1. Introduction

Today, with the development of deep learning techniques, the production of fake content has reached a rapid and highly realistic level. Deepfake images, videos, and audio content produced using Generative Adversarial Networks (GANs) and more recently diffusion-based models pose serious threats to security and society, and distinguishing reality from fiction is becoming increasingly difficult to the naked eye [1]. As of 2023, the number of deepfake video contents has tripled, while the number of speech-based deepfakes has increased eightfold [2,3]. This increase in deepfake production leads to serious security risks,

particularly on social media platforms, such as the rapid spread of misinformation, identity theft, blackmail, fake evidence production, and political manipulation. This situation has made it difficult to maintain accuracy in the digital environment and has highlighted the importance of fake content detection efforts [4,5]. However, despite significant progress in detection methods, most existing approaches focus on a single modality, such as images, videos, or audio signals. Each modality presents distinct technical challenges. In image-based detection, modern generative models reduce visible artifacts, making it increasingly difficult to extract discriminative features. In video-based detection, effectively modeling both spatial and temporal dependencies remains a

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significant challenge, particularly for subtle manipulations across frames. Recent studies based on hybrid Convolutional Neural Network (CNN)–Long Short-Term Memory (LSTM)–Transformer and vision transformer architectures have shown improved performance; however, generalization and robustness issues still persist [6,7]. In audio-based detection, environmental noise, compression artifacts, and variability in recording conditions significantly affect model robustness [8]. These modality-specific limitations reduce the effectiveness of unimodal systems in real-world scenarios. To address these challenges, recent studies have explored multimodal approaches, which integrate information from multiple data sources. While multimodal approaches improve detection performance, they introduce new challenges, particularly in feature fusion, cross-modal alignment, and model interpretability. Efficiently combining heterogeneous features remains a complex task, and improper fusion strategies may lead to information loss or model bias [9]. Beyond detection performance, the increasing complexity of deepfakes has also created a need for systems that can explain their decisions. These artificial intelligence models, referred to as black-box models, make the detection process incomprehensible to users. Particularly in fields such as security, forensic computing, and public policy, it should be understandable why AI models classify certain content as fake. Therefore, Explainable Artificial Intelligence (XAI) has become a critical tool for making the decision processes of deepfake detection models transparent [10–12]. Although several XAI techniques, such as Shapley Additive Explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), Layer-wise Relevance Propagation (LRP), and Gradient-weighted Class Activation Mapping (Grad-CAM), have been applied to deepfake detection, these methods are generally used independently and within single-modality frameworks. There is still a lack of a unified explainability design that can consistently operate across spatial, temporal, and feature dimensions in multimodal systems [13]. This gap limits the usability of existing systems in real-world applications where both accuracy and interpretability are required. Instead of limited approaches focused on a single media type, there is a need for integrated systems that encompass multiple media types and present decisions made with high accuracy in a transparent and interpretable manner. To address this gap, this study proposes a unified and explainable deepfake detection framework that reliably distinguishes between real and fake content across different media types while providing interpretable insights into the detection process.

Despite the significant progress in deepfake detection, several critical gaps remain in the literature. First, most existing studies focus on unimodal data, limiting their ability to capture inconsistencies across different media types. Second, current multimodal approaches often rely on simple fusion strategies and fail to effectively model cross-modal relationships. Third, XAI techniques are typically applied in a fragmented manner and are not integrated into a unified framework that can provide consistent interpretations across modalities. Finally, there is a lack of user-oriented systems that combine high detection performance with interpretable and accessible outputs. These limitations highlight the need for a unified, multimodal, and explainable deepfake detection framework. To address these shortcomings, this study utilized separately optimized models, achieving accuracy rates of 99% for images, 96% for videos, 98% for audio, and 99% for multimodal data. The most notable aspect of the system is that it not only detects fakes but also explains in detail how decisions are made using powerful XAI techniques such as SHAP, LIME, LRP, and Grad-CAM. These explanations go beyond visual outputs and are supported by textual comments that users can understand. All these processes are combined with the Gradio interface to create a comprehensive and accessible deepfake analysis platform suitable for real-world use. In this way, the study presents a unique system that provides high accuracy and high explainability by addressing multiple media types with a holistic approach.

The main contributions of this study are summarized as follows:

1. A novel unified architecture is proposed that simultaneously processes image, video, audio, and audio-visual multimodal data within a single framework, enabling consistent analysis across different data modalities.
2. A lightweight cross-attention mechanism is introduced to effectively align and fuse heterogeneous audio and visual features, improving the detection of cross-modal inconsistencies in deepfake content.
3. Different XAI techniques (SHAP, LIME, LRP, Grad-CAM) are systematically adapted to each modality based on their structural characteristics, enabling more meaningful and context-aware explanations.
4. A standardized explainability pipeline is developed to generate both visual and textual explanations across all modalities, ensuring interpretability at spatial, temporal, and feature levels.
5. All detection and explainability components are integrated into a user-friendly interface using Gradio, enabling real-time analysis and practical usability in forensic and applied scenarios.

The remainder of the article continues as follows: [Section 2](#) details the relevant studies. [Section 3](#) covers methodology and methods. Here, the models used for four separate models are explained in detail. [Section 4](#) analyses the experimental results, evaluation parameters, and outputs obtained using explainable artificial intelligence. Finally, [Section 5](#) presents the conclusions.

2. Related work

With the increasing use of deepfakes, numerous studies have been conducted on their detection. This article explains some of these studies, classified by media type.

2.1. Deepfake image

In a study conducted by Mohan and colleagues [14], a CNN model based on DenseNet121 was developed to distinguish between real and fake images, and the model's accuracy rate was reported to be 93.42%. Furthermore, to explain the model's decision-making process, regions effective in classification were visualized using the Grad-CAM method. In this study, a method combining a specially designed CNN architecture with a multi-head self-attention (MHSA) mechanism was proposed for detecting deepfake images [15]. The model, tested on their own CCDDF dataset and popular datasets (140 K Real and Fake Faces, Celeb-DF v2), achieved 97–98% accuracy and AUC values above 99%. Furthermore, the model's decision processes were visualized using attention heatmaps, increasing its interpretability. In a study by Kumar et al. [16], deepfake detection was performed using a deep learning-based ensemble model approach. The model combined predictions from CNN architectures such as VGG, Xception, RegularisedConvNet, and RegularisedConvDenseNet, achieving 94.5% accuracy and an AUC value of 0.98. LIME, an explainable artificial intelligence method, was used to reveal which facial features the model focused on. In another study [17], the deep learning-based CNN models InceptionResNetV2, DenseNet201, InceptionV3, and ResNet152V2 were processed separately and the results were compared. The highest accuracy rate of 99.87% was achieved with the InceptionResNetV2 model. The model's reliability and decision mechanism were displayed in percentage segments using the LIME algorithm. In this experiment conducted by Jheelan and colleagues [18], different deep learning models—ResNet50, DenseNet121, MobileNet, InceptionV3, and a custom CNN—were compared to detect fake face images generated by StyleGAN. The highest accuracy rate of 98.5% was achieved with the MobileNet model. The study also developed a Flask-based web interface, enabling users to upload images and perform real/fake analysis.

Despite the high accuracy achieved in these studies, several limitations remain. Most image-based approaches rely heavily on spatial features and fail to generalize well against increasingly sophisticated

generative models, particularly diffusion-based methods. Additionally, although XAI techniques such as Grad-CAM and LIME are frequently applied, they are often used as post-hoc visualization tools rather than being integrated into the model design, limiting their effectiveness in providing consistent and reliable explanations. Furthermore, these studies focus exclusively on unimodal data, ignoring the complementary information available in other modalities. This highlights a key research gap in developing generalizable, multimodal, and inherently explainable detection systems.

In our study, the InceptionV3 model was used to detect deepfake images, achieving 99% accuracy and a 99% AUC value. To explain the model's decision-making processes, Saliency and LIME methods were applied, thereby regionally identifying fake facial areas and making them interpretable. What distinguishes this study from similar ones in the literature is that it not only offers high accuracy and visual interpretability but also makes the system open to user interaction by developing a Gradio-based user interface. Another difference is that this interface provides a high level of transparency and interpretability by supporting model outputs with both visual and textual explanations.

2.2. Deepfake video

Samuel Henrique Silva [10] proposed an explainable forensic analysis approach for deepfake detection. In the study, a hierarchical ensemble architecture consisting of attention-based deep learning models was used, and the model's decision regions were visualized using Grad-CAM. The system achieved 92.2% accuracy on the DFDC dataset and maintained its generalisability on different datasets. Konstantinos Tsigos [19] proposed a new framework to evaluate the performance of explainable artificial intelligence methods in deepfake detection models. This method measures how accurately explanation algorithms can identify the most influential regions on the decision in fake images, relative to the accuracy drop caused by manipulating these regions with adversarial attacks. The study compared a deepfake detection model trained on the FaceForensics++ dataset with five different XAI methods, and the LIME method emerged as the most successful explanation tool. In this study [20], the authors propose a deep feature stacking architecture that combines features extracted from multiple deep learning models for deepfake detection and a *meta*-learning-based classifier that operates on these stacked features. The study utilised Xception and EfficientNet-B7 models as base learners, and a feature-selection process was applied to combine the deep features obtained from these models and select the most discriminative features. The proposed model achieved 96.33% accuracy on the Celeb-DF (V2) dataset and 98% accuracy on the FaceForensics++ dataset. The explainable AI aspect was not fully focused on; it was only demonstrated using Grad-CAM and heatmaps after the Xception and EfficientNet-B7 models. In this study by Ishrak [21], frame-by-frame analysis is performed by combining CNN and CapsuleNet architectures with LSTM to detect deepfake videos. An accuracy rate of 88% was achieved on the DFDC dataset. Grad-CAM is used to analyse and display the reasons behind the model's decisions. Malolan [22] trained the Xception model on face frames extracted from the FaceForensics DeepFakeDetection dataset for deepfake video detection. An accuracy rate of 94.33% was achieved. Furthermore, Explainable AI techniques such as LRP and LIME were applied to explain the model's decisions and increase its reliability, and the face regions the model focused on were visually represented.

Although these studies demonstrate strong performance in video-based deepfake detection, several critical challenges persist. Most approaches rely on frame-level analysis or limited temporal modeling, which restricts their ability to capture long-range dependencies and subtle manipulations. Moreover, explainability is typically limited to visual attention maps, which do not fully explain temporal decision dynamics. Another key limitation is the lack of integration with other modalities, particularly audio, which plays a crucial role in real-world deepfake scenarios. These limitations indicate that current methods

are insufficient for holistic and interpretable analysis of dynamic manipulations, necessitating multimodal and temporally-aware explainable frameworks.

In this study, we developed a two-stage deep learning model that operates at both the frame and video sequence levels. In the first stage, powerful features with high representational power were extracted from the videos. In the second stage, these features were processed, taking into account the temporal context, to create a hybrid model. This model enabled effective deepfake detection at both the frame level and the video sequence level. In frame analysis, an accuracy rate of 96.76% and an AUC rate of 99.38% were achieved. With the hybrid model, these rates were 90.88% accuracy and 98.06% AUC. A key aspect distinguishing this study from others is that Grad-CAM was applied not just to a single frame but to multiple frames, providing more comprehensive and consistent explainability. This enables the reliable detection of manipulations occurring even in very brief moments within a video.

2.3. Deepfake audio

In this study by Bisogni [23] and colleagues, a lightweight machine learning framework targeting audio-based deepfake detection is presented. This model also distinguishes between real and fake audio recordings by utilising spectral, temporal, chroma, and frequency-level features. In experiments conducted on the ASVspoof2019, FakeAVCelebV2, and In-The-Wild datasets, the study achieved accuracy rates of 89%, 94.5%, and 94.67%, respectively. The model's decision mechanism was analysed using SHAP, an explainability technique, critical features for deepfake audio detection were identified, and the model's transparency was increased. In another study [24], fake voices were generated using GAN with the FoR dataset, and quality assessment was performed using the Fréchet Audio Distance metric. In deepfake voice detection, the model's decision mechanism was analysed using Explainable AI methods such as LIME, SHAP, and Grad-CAM. In this study by Lim [25], explainable deep learning techniques are used for deepfake voice detection. In the study, deepfake voice detection was performed using mel-spectrogram-based CNN, CNN-LSTM, and permutation CNN-LSTM models, and decision processes were made explainable using explainable artificial intelligence techniques such as Integrated Gradients, LRP, and Deep Taylor. In this study conducted by Channing [26] and his team, voice deepfake detection was performed using models such as AST, Wav2Vec, and GBDT, and attention and occlusion techniques were applied to explain the decision-making processes. In another study by Ning Yu [27] and colleagues, a feature fusion method combining raw waveforms and spectrograms is proposed for speech deepfake detection. Using the SHAP method to explain the model's decisions, the most critical features distinguishing fake from real speech were visualized using heat maps.

Despite promising results, audio-based deepfake detection methods face several challenges. Most approaches depend on handcrafted or spectral features, which are sensitive to noise and environmental variations, limiting their robustness in real-world conditions. While XAI methods such as SHAP and LIME provide feature-level interpretability, they often fail to capture temporal dependencies and interactions between frequency components. Additionally, these methods are typically applied independently and are not integrated into a unified framework. Another major limitation is the absence of cross-modal validation, as audio-only systems cannot verify inconsistencies with visual data. This reveals a significant gap in developing robust, multimodal, and temporally explainable audio detection systems.

Our study also achieved explainability using two systems without looking at more than one feature. The XGboost model was used to focus on the important features of the voice, and SHAP and LIME were used to explain these features by showing their importance scores. Furthermore, using CNN + LSTM, analysis was performed temporally based on frequency, achieving an accuracy of 99%. Explanations for the frequencies were provided using Grad-CAM and LRP models. Unlike other studies,

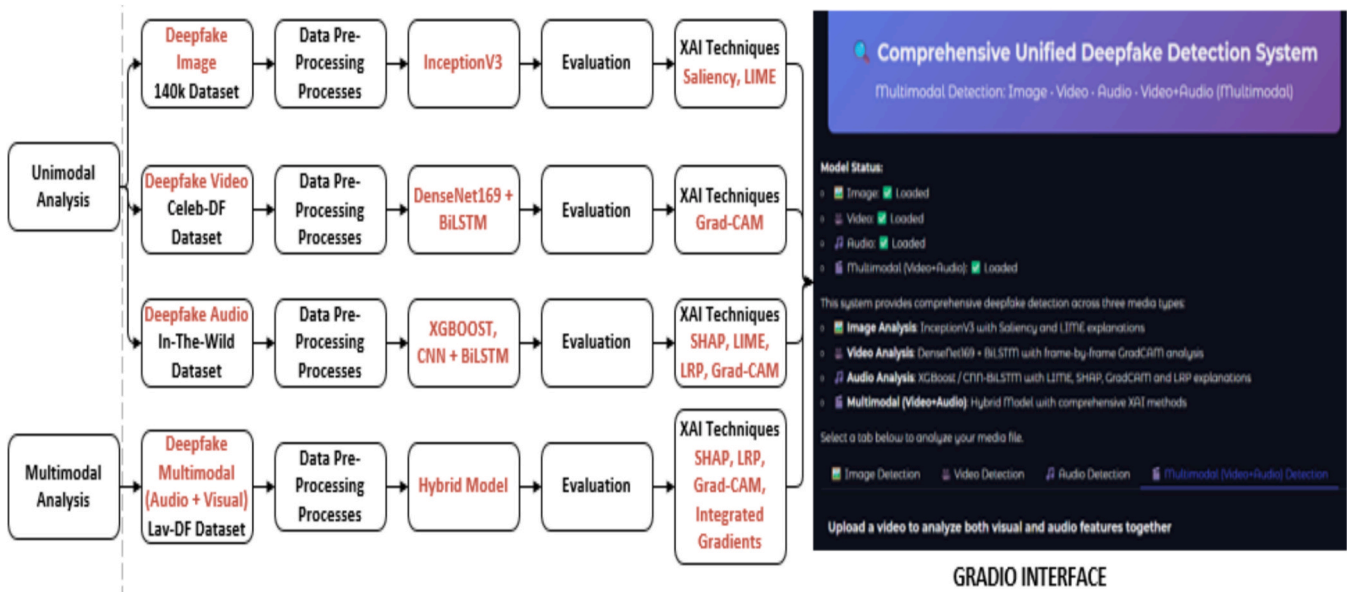


Fig. 1. General workflow diagram.

not only was a high accuracy achieved, but more explanations were also provided, creating a user-based comprehensive system with the Gradio interface.

2.4. Deepfake multimodal (Audio + Visual)

In the study presented by Cai et al. [28], a large-scale, localisation-focused dataset called LAV-DF is proposed for audio-visual forgery detection. The authors introduced the BA-TFD and improved BA-TFD + models for detecting fake segment boundaries on this dataset and presented that the BA-TFD + model achieved a performance of 96% at the AP@0.5 level. Javed et al. [29] used LAV-DF and other datasets in their study. In this work, local visual features (e.g., eye and lip movements) were extracted using CNN, while global contextual relationships were modelled using Vision Transformer. They achieved an accuracy rate of 98.12% on the LAV-DF dataset. In this study by Park et al. [30], multimodal fake video detection is performed using the probabilistic score level fusion of two powerful unimodal models that take sound and image modalities in their raw form. The X3A model performed with an accuracy of 99.60% / AUC = 0.9999 on the AV-Deepfake1M benchmark. Cheng et al. [31] proposed CLFormer, a cross-language transformer-based model that combines audio and visual modalities to perform time-based forgery localisation. CLFormer achieved an accuracy rate of 99.33% AP at tIoU = 0.5 on LAV-DF, proving successful in a cross-language scenario. In this study presented by Rahman et al. [32], a multimodal framework combining visual (EfficientNetB0) and auditory (Mel-Frequency Cepstral Coefficients (MFCC) + BiLSTM) features for fraud detection is proposed. This model achieved an accuracy rate of 87.25% on the AVLips dataset. In terms of explainability, attention maps reveal the critical regions of the face on the visual side, while on the audio side, they reveal which MFCC components are decisive. ERF BA TFD+, proposed by Zhang et al. [33], is a model developed for audio-visual multimodal deepfake detection and localisation. By adding the Evidence-based Reasoning Framework (ERF) to the BA TFD architecture, the model's decisions can be interpreted through attention weights and reconstruction errors. In the visual modality, 3D-CNN-based features were used, while in the audio modality, MFCC-based temporal features were employed. Thanks to multi-scale and cross-modal attention mechanisms, both the detection and localisation of false segments were achieved. ERF BA TFD + demonstrated performance with values of AP@0.5 = 0.9630 and AP@0.75 = 0.8498 on the LAV-DF dataset.

Although multimodal approaches demonstrate high performance in deepfake detection, they still suffer from significant performance degradation in the effective fusion of heterogeneous features and synchronization inconsistencies between audio-visual streams. Most existing studies rely on simple fusion strategies that neglect cross-modal interactions, leaving them inadequate against asynchronous data in real-world scenarios. Furthermore, the most prominent deficiency in terms of explainability (XAI) is that techniques such as Grad-CAM or SHAP are generally applied independently and in a fragmented manner for each modality. This prevents the provision of a holistic and coherent interpretation reflecting cross-modal interactions, which is particularly critical in high-stakes fields like digital forensics. Consequently, there is a clear need in the literature for integrated systems that can simultaneously address feature fusion, alignment, and explainability within a unified framework.

In our study, an integrated multimodal architecture capable of extracting video and audio simultaneously was developed. The model evaluates the relationships between audio-visual features in a common representation space, enabling consistent analysis of both visual and auditory forgeries. This model demonstrated robust performance, achieving 96.32% accuracy and 99.68% AUC. While explainability was limited in previous studies, this model offers a detailed explainability process for both audio and visual modalities. By analysing the consistency between video and audio, the source of the forgery is revealed more clearly. As with our other models, a user-friendly Gradio interface was developed for this study, creating a comprehensive system. All these features distinguish this system from existing multimodal approaches in the literature.

3. Materials and methodology

This study proposes a comprehensive deepfake analysis framework that evaluates various media types through both unimodal and multimodal learning strategies. Unlike fully integrated multimodal systems, the proposed approach is designed to analyze each modality independently while also incorporating a dedicated multimodal model for joint audio-visual analysis. The overall workflow of the framework is illustrated in Fig. 1. As shown in the figure, the framework consists of two main components:

Unimodal Analysis: A stage where image, video, and audio data are processed separately using modality-specific models.

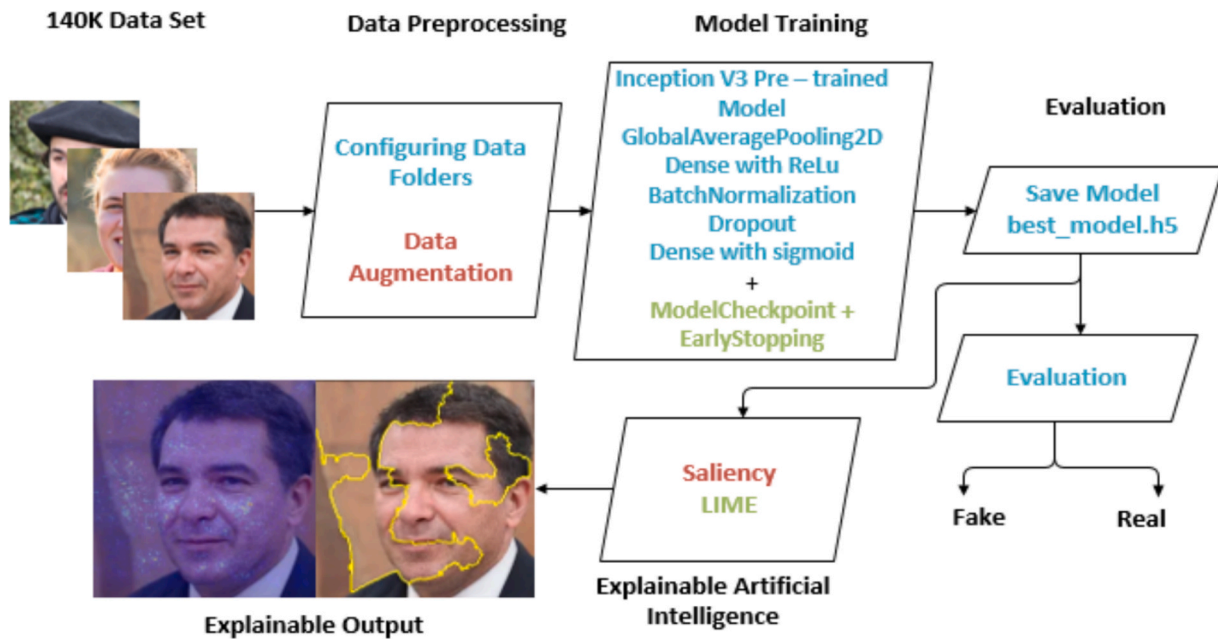


Fig. 2. Deepfake image workflow diagram.

Multimodal Analysis: A stage where audio and visual information are jointly analyzed within a specialized fusion architecture.

In the unimodal stage, each data type is processed independently to preserve its intrinsic characteristics and maximize detection performance. Specifically, deep learning models optimized for each modality are trained separately, and XAI techniques are applied to interpret their decisions. This design ensures that each modality is analyzed in detail without being affected by the inherent limitations of other data types. In contrast, the multimodal stage is designed to capture cross-modal inconsistencies by jointly analyzing audio and visual data. Rather than relying on simple feature concatenation, a cross-attention-based mechanism is employed to model the interaction between modalities. This enables the detection of complex deepfake patterns that may not be observable through unimodal analysis alone.

Finally, all analysis results for each model are presented through a user-oriented interface, providing both predictions and explainability (XAI) outputs in a structured and interpretable manner. Each specific component will be described in detail in the subsequent sections.

Overall, the selected models (InceptionV3, DenseNet169, BiLSTM, CNN, XGBoost) were specifically chosen to align with the structural and temporal characteristics of each modality, providing superior performance compared to traditional approaches. All experiments were conducted on a system equipped with an NVIDIA GPU (GTX 1050) and 8 GB of RAM. TensorFlow's dynamic GPU memory growth feature was enabled to optimize memory management. The software environment consisted of Python 3.x, TensorFlow 2.x, OpenCV, Librosa, scikit-learn, and XAI libraries, including SHAP, LIME, and Grad-CAM. To ensure reproducibility of the results, a fixed random seed of 42 was assigned to all random number generators in NumPy, TensorFlow, and Python's built-in modules.

3.1. Deepfake image

In this study, operations were primarily performed on deepfake images. After pre-processing the dataset, the data was trained using the InceptionV3 model and evaluated using evaluation parameters. Subsequently, the images were interpreted and made more transparent using Saliency and LIME, which are techniques from Explainable Artificial Intelligence, along with the obtained model file. Fig. 2 shows the workflow diagram for the images.

3.1.1. The dataset used for deepfake image

A dataset of 140,000 images, accessible via Kaggle and consisting of real and synthetic faces, was used for deepfake visual detection. Half of the images in this dataset are real human faces, while the other half are synthetic faces generated using the StyleGAN model. The real faces were taken from the FFHQ collection, while the fake faces were taken from a large-scale fake face dataset based on StyleGAN. The images obtained from both sources were combined and reduced to a resolution of 256x256 pixels. These were organised into training, validation and test subsets for the study [34].

3.1.2. Data pre-processing process for deepfake image

To achieve a higher accuracy rate, certain pre-processing steps were undertaken. Firstly, to ensure consistency in model training, the random seed value was fixed for NumPy, TensorFlow, and Python. The dataset was divided into training, validation, and test subsets while preserving the original directory structure, with each subset organised under the 'real' and 'fake' classes. The directories were scanned to verify the number of images in each subclass and to check the data structure. The training folder contains 50,000 real and 50,000 fake images, the validation folder contains 10,000 real and 10,000 fake images, and the test folder contains 10,000 real and 10,000 fake images. All images were scaled to 256x256 pixels to match the model input size. Data augmentation was applied to the training data using rotation = 20, scaling = 0.2, horizontal flipping, and fill_mode = nearest parameters, aiming to make the model more robust against different geometric variations. In the validation and test datasets, only the normalisation process specific to the InceptionV3 architecture was performed to ensure unbiased measurement of model performance, and no augmentation was applied. Thus, all prepared data were made ready for transfer to the model.

3.1.3. Recommended model for deepfake image

Inception-v3 is an architecture designed to increase computational efficiency and improve representational power in deep convolutional neural networks. The model reduces the number of parameters and increases learnability by decomposing large filters into smaller components [35]. The factorised convolution approach at the core of Inception-v3 aims to reduce computational cost while maintaining performance. For example, converting 5×5 convolutions into two consecutive 3×3 convolutions reduces parameters by approximately 28% [30,31]. This

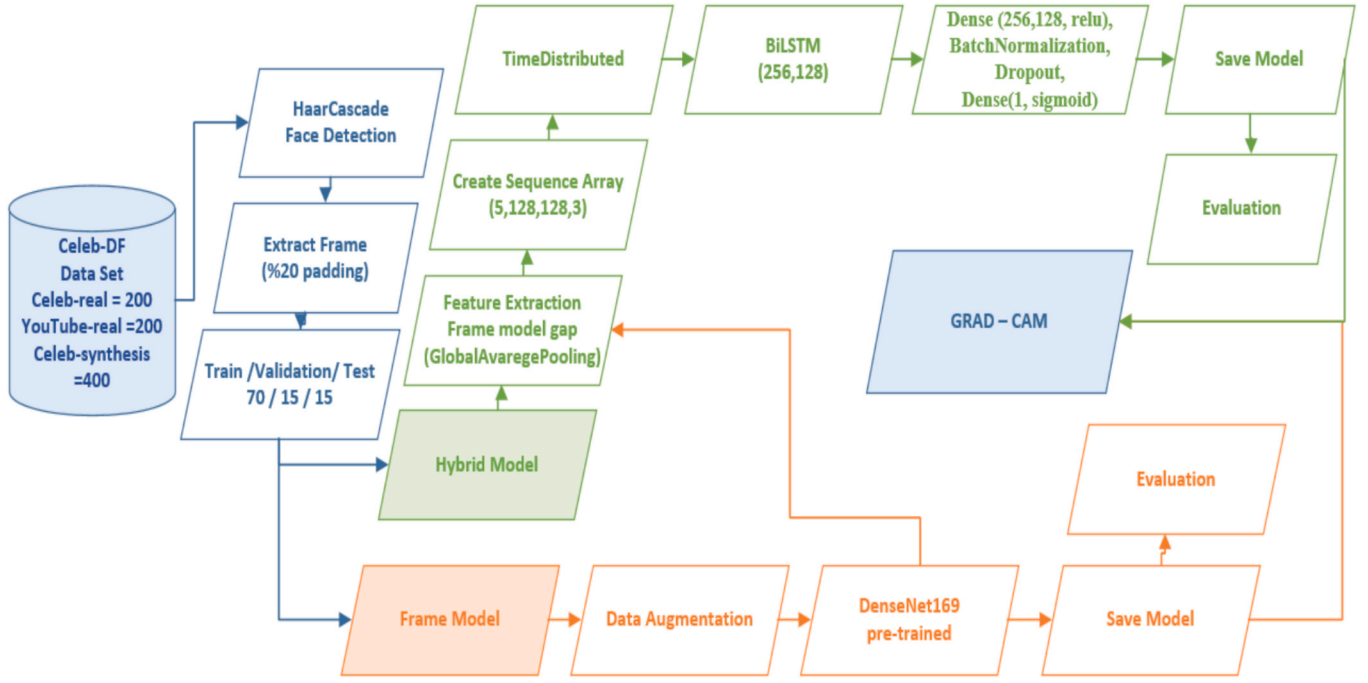


Fig. 3. Deepfake video workflow diagram.

multi-scale and modular structure enables the parallel operation of filters of different sizes, allowing for the simultaneous learning of broad and deep features. Consequently, InceptionV3 offers a strong representation learning capacity, particularly in capturing subtle texture differences, light transitions, and artefacts specific to synthetic face generation in facial images. For this reason, this architecture was selected for our study.

Transfer learning was performed on the InceptionV3 model using weights previously trained on ImageNet. This method enables the transfer of learned image features in large datasets, while ensuring a much faster and more stable learning process when the dataset is smaller. When creating the model, the classification layer of InceptionV3 was removed, and only the basic structure for feature extraction was used. Approximately 50 layers from the first layers of the network were frozen to preserve low-level universal features. Subsequently, the model consists of GlobalAveragePooling, Dense(512, ReLU), Batch Normalisation, Dropout(0.5), and Dense(1, sigmoid) output layers. This structure contributes to both reducing overfitting and making the real-fake distinction more robust. It was compiled with the binary_crossentropy loss function and the Adam optimisation algorithm with a low learning rate (0.0001), as appropriate for a binary classification problem. During the training process, ModelCheckpoint was used to record the best performance, and the EarlyStopping (patience = 5) mechanism was used to terminate the training process when no progress was made in validation performance. This prevented unnecessary training cycles and ensured that the model with the best accuracy rates was recorded. Training was completed with both training and validation data streams in accordance with 15 epochs and a batch size of 32.

3.1.4. Recommended XAI models for deepfake image

To make the decision mechanism of the deep learning model developed within the scope of this study understandable, two different explainable artificial intelligence methods were applied: saliency maps and LIME. Both methods complement the model's prediction process, with one focusing on the model's internal sensitivity structure and the other on its external behaviour. Saliency evaluates the effect of each pixel in the input image of a deep learning model on the model output in a derivative-based manner. This method helps us calculate the pixels

and regions that contribute most to the decision process. The model's sensitivity to that pixel is measured by calculating the relationship between a small change in the output and a small perturbation in the input pixel [36].

In our study, saliency maps were obtained using GradientTape, TensorFlow's automatic differentiation mechanism. This process consists of the following stages: making the input tensor differentiable, forward passing the model, differentiating the output with respect to the input pixel tensor, performing channel-based summarisation, min-max normalisation, and overlaying the map onto the image. The hyperparameters used for saliency maps were: alpha = 0.4, colormap = Jet, adaptive threshold = max_score × 0.3, minimum threshold = 0.01, and top-k = 3. As a result of this process, it becomes possible to visualise the extent to which the model is sensitive to specific anatomical regions of the face (eyes, nose, lip area, jawline, etc.).

Another technique used, the LIME method, is a model-independent approach that generates explanations at the example level. Its basic principle is to divide the image into superpixel segments and measure how the model responds to given perturbations by masking these segments with different combinations. LIME approximates this relationship through a local linear model and produces an importance coefficient for each segment [37]. In this study, the image was first divided into superpixel clusters, then 300 perturbed examples were generated, considering 10 features and top 2 labels, and their model outputs were recorded. The hyperparameters for LIME were: hide_color = 0, random_seed = 42, adaptive threshold = max_score × 0.2, minimum threshold = 0.005, and top-k = 3. Subsequently, a local linear model was trained to identify the segments with positive contributions and the highest weights; these segments were then annotated on the original image. Unlike the saliency approach, LIME relies on the model's externally observable behaviour rather than its internal derivative information.

The regional importance maps obtained from both methods were matched using facial anatomy references such as the forehead, eye area, nose, cheeks, lip line, chin, and hair/headline. This alignment enables a detailed assessment of which facial regions the model focuses on when making decisions. In the final stage, Saliency analysis, LIME outputs, model predictions, textual explanations, and confidence scores were

combined to form a consistent technical explanation.

3.2. Deepfake video

After the images were identified, operations were performed on the deepfake video. After pre-processing the dataset, the data was trained using the frame model and hybrid model and evaluated using evaluation parameters. Subsequently, deepfake analysis can be performed on multiple frames using Grad-CAM, one of the Explainable Artificial Intelligence techniques, along with the obtained model file. Fig. 3 shows the workflow diagram created for the video.

3.2.1. The dataset used for deepfake video

In this study, the Celeb-DFV2 dataset was used to determine whether video content was real or fake. This dataset is a comprehensive video collection created for the purpose of developing deepfake detection systems. The dataset contains 5,639 high-quality fake videos derived from 300 videos taken from YouTube and 590 real celebrity videos created in the celeb-real file. The videos feature variable lighting, different camera angles, various facial expressions, and characteristics such as different ages, genders, and ethnicities. This diversity makes it possible to test the generalisation capacity of the developed algorithms [38].

3.2.2. Data pre-processing process for deepfake video

A comprehensive data pre-processing process was carried out to make the video-based facial data used in this study suitable for model training. First, to ensure the reproducibility of experimental results and reduce variance caused by randomness, a common seed operation was performed on the built-in randomness modules of NumPy, TensorFlow, and Python. Subsequently, since raw video data was not processed directly, frame extraction was applied at specific sampling intervals from the videos to reduce computational costs and keep the data volume under control. In each video stream read via OpenCV, frames were sequentially scanned, and only frames corresponding to the sampling rate were recorded using a thresholding method based on the frame index. This approach ensured the dataset was thinned in the time dimension and created a more balanced sample structure. The pre-trained face detection model in OpenCV, based on Viola's [39] proposed Haar Cascades classifier, was used for face detection in the obtained frames. Each frame was first converted to greyscale, then face regions were searched for at different scale levels. If more than one face was detected, the largest face region was accepted as the dominant face in the frame and processed. After removing the face region, approximately 20% additional contextual area was added around the bounding box, thus ensuring the preservation of complementary anatomical elements such as the hairline and jawline located within the natural boundaries of the face. The facial region was then normalised in size, and all data was reduced to a resolution of 128×128 pixels to achieve a homogeneous distribution. In the final stage of the pre-processing, the obtained facial images were divided into 70% training, 15% validation, and 15% test subsets. Data augmentation was also added to the training data for the frame model. `rotation_range = 20`, `width shift range = 0.2`, `height shift range = 0.2`, `horizontal flip = True`, `zoom range = 0.15`, `fill mode="nearest"` were set. One video is generally between 10–15 s long. One frame is taken from every 40 frames to extract the face and frame. For the hybrid model, each sequence must contain 5 frames, and the stride number is set to 2, meaning the window will advance by 2 frames each time.

3.2.3. Recommended model for deepfake video

After preprocessing, the data has been prepared for the model. DenseNet is a CNN architecture that increases depth, enhances feature propagation, and reduces the number of parameters by directly connecting each layer to the outputs of all previous layers. These dense connections reduce the vanishing gradient problem while also

supporting feature reuse. DenseNet-169 was chosen from different depth options due to its performance-computational cost balance. DenseNet-169 consists of four dense blocks; each block contains 1×1 and 3×3 convolution layers. The transition layers at the end of the blocks optimise information flow within the network by adjusting the size of the feature maps [40].

Another model, BiLSTM, is an advanced variation of the classic LSTM [41] architecture that enables it to learn temporal dependencies in both directions. While traditional LSTMs only allow information flow from the past to the future, BiLSTM architectures learn the entire representation of temporal data by using two separate LSTM layers that operate in both forward and backward directions.

We developed a two-stage architecture that can learn at both the frame and video sequence levels for deepfake video detection. In the first stage, to perform robust feature extraction at the frame level, the classification layers of the pre-trained DenseNet169 architecture on ImageNet were removed, and all layers except the last 30 layers were frozen, using only the convolutional body. This reduced computational cost while enabling the learning of high-level representations through fine-tuning. The feature maps obtained were then fed into a Global-AveragePooling layer, followed by Dense (512, ReLU), BatchNormalisation, Dropout(0.4), Dense(256, ReLU), Dropout(0.3), and Dense(1, sigmoid). Low-learning-rate (0.0001), Adam optimisation was employed, with binary_crossentropy chosen as the loss value.

In the second stage, to model temporal relationships at the video level, the previous DenseNet169-based architecture was used as a feature extractor up to the GAP layer, and only the last 15 layers were fine-tuned. Subsequently, the TimeDistributed structure was applied to each frame. The multidimensional feature sequences obtained from the video sequences were processed in two-stage BiLSTM layers (256 and 128 units) that could learn temporal dependencies from both past and future contexts. This structure ensured that facial movements, expressions, micro-expressions, and inconsistencies specific to synthetic generation were effectively captured in the temporal dimension. Following the BiLSTM layers, generalisation performance was enhanced by adding Dense (256, ReLU), BatchNormalisation, dropout(0.4), Dense (128, ReLU), and dropout(0.3). The output layer was configured with sigmoid activation for video-based binary classification. This enabled analysis both at the frame level and temporally. Both models were trained for 25 epochs using the ModelCheckpoint, EarlyStopping(patience = 5), and ReduceLROnPlateau (patience = 3, min_lr = $1e-7$, factor = 0.5) parameters.

3.2.4. Recommended XAI models for deepfake video

GradCAM (Gradient-weighted Class Activation Mapping), one of the explainable artificial intelligence techniques, aims to visualise which spatial regions play a decisive role in the decision-making processes of convolutional neural networks [42]. GradCAM calculates the gradients of the prediction output belonging to a specific class on the activation maps of the model's final convolutional layer, takes the channel-based average of these gradients, and multiplies the resulting weights by the relevant activation channels to produce a class-specific importance map [42]. In the code, this process is achieved by identifying the model's final convolution layer, using the GradientTape mechanism to automatically calculate the derivative between this layer's output and the model prediction, and subjecting each channel's gradients to global average pooling. The obtained channel weights are reflected in the spatial dimensions of the activation map to produce pixel-level importance values. Negative values are then zeroed, ReLU is applied, and normalisation is performed to bring the values into the 0–1 range. This 2D importance map is scaled to the same size as the input face image and coloured using a thermal colour map, visually highlighting the facial regions that contribute most to the model's decision.

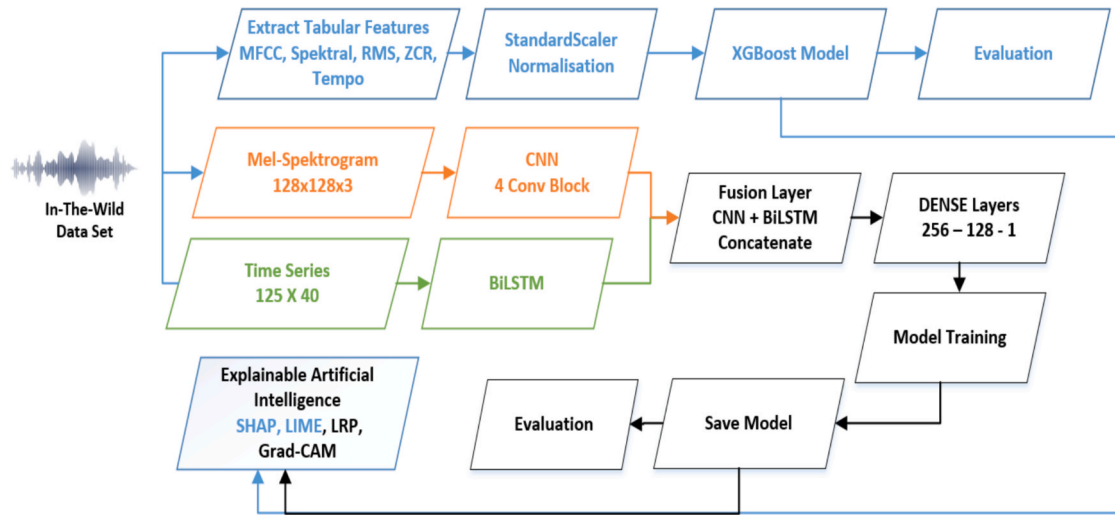


Fig. 4. Deepfake audio workflow diagram.

3.3. Deepfake audio

Another type of media we use is audio. First, the data pre-processing process was carried out. Afterwards, different pre-processing processes were performed for the audio files, and models were developed according to their features. XGBoost was used for tabular features, while a CNN + BiLSTM hybrid model was used for mel-spectrogram and MFCC features. After these training processes were completed, an evaluation was performed. For explainability, SHAP, LIME, LRP, and Grad-CAM were used to add comprehensive interpretability and transparency. The workflow diagram for the processes performed for audio is shown in Fig. 4.

3.3.1. The dataset used for deepfake audio

An in-the-wild dataset was used for deepfake voice detection. The dataset consists of a total of 37.9 h of recordings compiled from the voices of 58 well-known individuals, including politicians and celebrities. In terms of content, it includes both fake and real voice samples recorded in natural environments, featuring noisy and variable recording conditions. It contains 20.7 h of real voice and 17.2 h of fake voice [43].

3.3.2. Data pre-processing process for deepfake audio

In this study, a multi-dimensional data pre-processing process was applied to make the audio data modelable. First, all audio samples were resampled to 16 kHz to ensure consistency in the calculations, and each file was trimmed to a fixed duration of 10 s. Subsequently, in the feature extraction phase, both summary features to be used in tabular format and temporal and visual representations (mel-spectrograms) for deep learning models were obtained. Within the tabular features, 40 MFCC features, 20 spectral statistics (mean, std, median, min, max, etc.), spectral features (centroid, bandwidth, rolloff), RMS (sound energy), ZCR (Zero Crossing Rate), and tempo features were extracted. Furthermore, two separate data representations were created for deep neural networks: Mel-spectrograms were resampled to 128×128 dimensions, normalised, and converted to RGB. MFCC sequences were also converted to a fixed-length sequence structure while preserving the time dimension (125 time steps, 40 features). StandardScaler was also applied for use only in tabular features. The dataset was split into training, validation, and test subsets at a ratio of 70–15–15, preserving the class distribution.

3.3.3. Recommended model for deepfake audio

To distinguish between fake and real voice samples, a hybrid

CNN-BiLSTM architecture has been developed that jointly processes Mel-spectrogram-based features and MFCC-based temporal information. The model comprehensively evaluates both the frequency components and the acoustic transitions over time in human speech. The architecture consists of two parallel components. The first model is a CNN branch containing four convolutional blocks (32, 64, 128, 256) to learn local frequency-textural patterns from Mel-spectrograms. This structure is supported by Batch Normalisation, ReLU, MaxPooling, and Dropout units and is converted into compact representations with Global Average Pooling in the final convolutional block. The other model consists of a BiLSTM branch (128,64 Units) + Dropout, comprising a two-layer bidirectional LSTM, which captures the forward-backward context of time series dynamics by modelling temporal dependencies in MFCC sequences. The outputs of both branches are concatenated and fed into Dense layers (256,128), followed by BatchNormalisation and a sigmoid activation output unit. The batch size is set to 32 and the epoch value to 50 to perform binary classification.

An XGBoost model was also used to analyse the tabular characteristics of the sound. The model parameters used were $n_estimators = 200$, $max_depth = 8$, $learning_rate = 0.05$, $subsample = 8$, $colsample_bytree = 0.05$, $gamma = 1$, $min_child_weight = 3$. Overfitting was prevented during model training by using the Early Stopping ($patience = 15$), $ReduceLROnPlateau$ ($factor = 0.5$, $patience = 7$, $min_lr = 1e-7$) and $ModelCheckpoint$ mechanisms.

3.3.4. Recommended XAI models for deepfake audio

After the models were trained, four different XAI techniques were used to explain not only the classification process into fake and real, but also the reasons behind these decisions and how they were made.

The first XAI technique used, SHAP, calculates each feature's contribution to the model based on Shapley values from cooperative game theory. By evaluating all possible feature combinations, it fairly distributes each feature's impact on the prediction, thereby making both the local and global levels of model operation and decisions understandable [44]. In the study, all trees of XGBoost are read through $TreeExplainer$, and the node split values of each tree, the output of each leaf, and all decision paths are analysed mathematically in polynomial time. Beeswarm visualisation is selected to calculate the overall effects on all data.

Another selected method is LIME [37]. It is a method that provides model-independent local explanations. Using $LimeTabularExplainer$, it creates a neighbouring data point by making random small changes around a selected example. It also produces hundreds of synthetic variations using tabular features. These variations are sent to the XGBoost

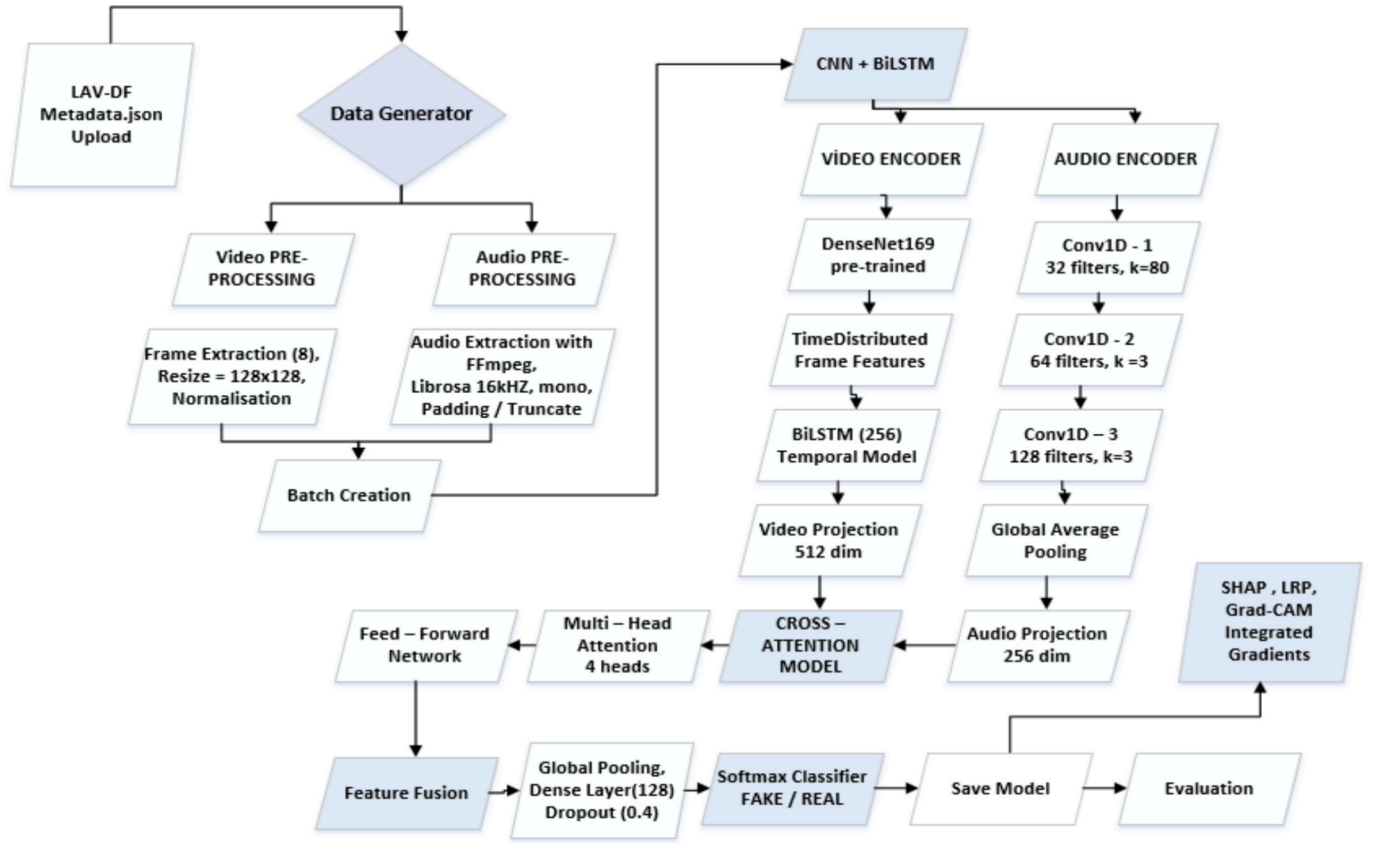


Fig. 5. Deepfake multimodal workflow diagram.

model, and the model's local behaviour is observed. LIME then weights the local examples to form a linear regression. The coefficients of the trained linear model can be displayed as a true/false bar chart, showing each feature's contribution to the The Grad-CAM [42] model was selected to provide interpretability for the CNN-BiLSTM model operating on the mel-spectrogram. Grad-CAM reveals which time–frequency regions the model focuses on by using the activations in the model's final convolutional layer. To do this, it calculates the gradients of the output score (fake/real) for the selected class relative to the model's final convolutional layer. These calculated gradients show how important each feature map in the convolutional layer is for the relevant class. These importance coefficients are then multiplied by the feature maps obtained from the convolutional layer to create a class activation map. This resulting heatmap visually and clearly shows which frequency bands and time intervals the model considers critical when making decisions on the mel-spectrogram. Thus, it becomes possible to understand which regions the CNN-BiLSTM model pays attention to when making its 'true' or 'false' classification.

Finally, the method used, LRP, is a technique that distributes the output of the deep learning model backwards through the layers to calculate the contribution of each input unit to the classification decision [45]. In this study, the Gradient x Input [46] approach, which is a simplified gradient-based LRP method, was used. It provides faster computation compared to the full LRP rules. Heat maps are generated at the pixel level on the Mel-spectrogram for sound. The formula used is:

$$\text{Relevance} \approx |\partial f / \partial x| \times |x|.$$

Here, the model output (f) measures how sensitive it is to each component (x) in the input. In this way, the sensitivity of f to each time–frequency domain in the mel-spectrogram is measured, then the energy amount of this cell is measured, and the two values are multiplied to obtain a relevance value. The relevance value indicates that the region that most significantly alters the model's output and

simultaneously carries the most information in the input is also the most important area in terms of decision-making.

3.4. Deepfake multimodal (Audio + Visual)

In addition to unimodal analysis, a dedicated multimodal model is developed to jointly analyse audio and visual information. Unlike the unimodal approaches, this model is specifically designed to capture cross-modal relationships and detect inconsistencies between audio and visual streams.

As illustrated in Fig. 5, the multimodal framework integrates features extracted from both video and audio encoders using a cross-attention mechanism. This approach allows the model to learn interactions between modalities rather than treating them independently. Consequently, the system is capable of identifying more subtle and complex deepfake manipulations that may not be detectable using unimodal models alone.

3.4.1. The dataset used for deepfake multimodal

LAV-DF is a large-scale multimodal deepfake dataset containing content-focused manipulations in audio, video, and audio–video combinations. By including examples where only selected short segments, rather than the entire video, are manipulated, it provides a realistic evaluation environment for both localising the timing of forgeries and temporal detection tasks. The dataset consists of a total of 136,304 videos belonging to 153 different identities. Of these, 36,431 are real, while 99,873 contain manipulated segments [28].

3.4.2. Data pre-processing process for deepfake multimodal

First, a custom data generator based on TensorFlow was designed, and each example in the dataset was processed with both visual and audio features. First, the json file containing the structural information

of the dataset was loaded, and real or fake labels for each video were extracted using the `modify_video`, `modify_audio`, `n_fakes`, and `fake_periods` parameters specified in the json file. This approach ensured data integrity by correctly labelling videos containing fake segments. The dataset was created with a balanced mix of real and fake examples, and any potential imbalances in class distribution were addressed by selecting an equal number of examples from each class. Subsequently, for each video, up to eight frames were extracted using equal-interval sampling from the visual stream, converted to the RGB colour space, resized (128 × 128 pixels), and normalised to the 0–1 range. This process ensures that a fixed-size model representation is obtained from videos of different lengths. For audio preprocessing, the audio from each video was extracted using FFmpeg in a single-channel, 16 kHz sampled WAV format. The resulting audio signal was read using librosa, trimmed to a fixed duration, or padded to complete it. This ensured that the model was trained with audio input of equal size across all samples. Any NaN or infinite values that may occur in the audio signal were checked and replaced with a zero vector where necessary. Any NaN or infinite values that may occur in the audio signal were checked and replaced with a zero vector where necessary. In the final stage, both visual and audio inputs were converted into numpy tensors to make them compatible with the model's input format. Additionally, at the end of each epoch, the sample indices were reshuffled to prevent the model from overfitting during the training process. For memory management, a total of 40,000 videos were taken from the training data, 16,000 from the validation data, and 6,000 from the test data.

3.4.3. Recommended model for deepfake multimodal

An integrated multi-modal deep learning architecture capable of extracting synchronous information from both video and audio has been designed to detect deepfake videos. The model consists of three main components: a video encoder, an audio encoder, and a cross-modal attention module.

The video encoder utilises the pre-trained DenseNet169 [40] architecture on ImageNet to analyse each frame at the spatial level. The first half of the model is frozen, while the other half is trainable. As the video stream is processed frame-by-frame using a TimeDistributed structure, spatial features are then modelled temporally using a bidirectional LSTM layer (256). This captures both the content representations of the frames and the sequential relationships between frames.

The audio encoder is built on a CNN architecture consisting of three-stage 1D convolutional layers (32, 64, 128 filters). Each of these layers uses ReLU activation, BatchNormalisation, and MaxPooling in the first two Conv1D stages. Following the convolutional operations, GlobalAveragePooling2D was applied, and the entire time axis was summarised to form a single feature vector. It was then duplicated using RepeatVector (8) to align with the subsequent model. Finally, a Dense layer with TimeDistributed projects each time step independently into a 256-dimensional representation space, followed by Layer Normalisation and Dropout to enhance training stability.

A lightweight cross-modal attention mechanism [47] has been applied to convert video and audio features into a common representation. This mechanism is based on a multi-head attention module, where video representations are processed as queries (Q) and audio representations as keys (K) and values (V). The cross-attention operation is computed using the scaled dot-product attention formula [48]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This establishes a mutual relationship between audio-based anomalies and image-based forgeries, enabling the detection of inconsistencies between audio and image. The obtained fusion output is summarised using GlobalAveragePooling. Subsequently, a 128-neuron Dense layer, LayerNormalisation, and Dropout are added to further strengthen the model. In the final stage, a two-class softmax function is used to perform

the real–fake distinction.

The parameters selected for the model are as follows: epoch 10, batch size = 16, learning Rate = 2e-4, optimizer = Adam, loss = Sparse Categorical Crossentropy. Furthermore, modelCheckpoint, ReduceLROnPlateau (patience = 2, min_lr = 1e-7, factor = 0.3) and EarlyStopping (patience = 3) were added for training to improve the training process and ensure the best model was saved.

3.4.4. Recommended XAI models for deepfake multimodal

Analysing video and audio simultaneously results in the model producing only a single output of either ‘real’ or ‘fake,’ which is not sufficiently transparent. This classification decision, based on both visual and audio components, needs to be interpreted in greater detail to understand which information sources are influential and to what extent.

To this end, the Grad-CAM approach was used to reveal which visual areas the model focused on. In this method, conv5_block32_concat is evaluated to find the last convolutional layer of DenseNet169; if this is not found, the previous layers or convolutional layers within the TimeDistributed layer are evaluated. If none are found, gradients derived directly from the video input are used. Once the final convolutional layer is found, the process proceeds as follows: video frames are fed into the model, feature maps generated from the convolutional layer are obtained, and classification is performed to make predictions. The gradient is calculated by backpropagation based on the score of the predicted class. The average of these gradients is taken and converted into vectors. The feature maps are multiplied by these vectors to determine the importance scores. Since negative values are not effective in the decision process, they are zeroed out, and the resulting heat map is then normalised. For each frame, it is overlaid semi-transparently on the original image, thus revealing which areas the model considers.

The second method employed was Integrated Gradients. This is a gradient-based algorithm that generates model outputs based on input features without any knowledge of the model's internal structure [49]. In our study, the aim was to examine the model's decision along a linear path created between an ‘information-free reference (baseline)’ and the ‘actual input’. In this regard, the first step was to use a baseline consisting entirely of black frames for videos and a baseline that was completely zeroed (silent) for audio signals. IG creates a linear path between these two endpoints.

Interpolated Input(a) = Baseline + a*(Original Input– Baseline).

It is calculated using the formula. Here, the value of a (alpha) varies between 0 and 1 and is calculated in 30 different steps. An intermediate input is created for each a value. An interpolation line containing 30 steps was created, and at each step, an intermediate state was created where the input data contained slightly more information relative to the baseline. This intermediate input is fed into the model to calculate the gradient for the fake/real class. The gradient is calculated for each intermediate input, thereby answering the question of how much the model's fake/real prediction decreases/increases if this feature is slightly increased for each pixel or sound sample. After calculating the changing features at each point along the path, the average of these gradients is taken.

$$IG = (\text{OriginalInput} - \text{Baseline}) * \text{AverageGradient}$$

Importance scores are obtained. The explainability and interpretability of these scores have been enhanced by displaying them with a heat map.

Another explanation method, simplified LRP, which is the Gradient x Input method we used earlier, also provides explainability. As a first step, input is provided to the model and the class is predicted. After the predicted class is selected as the target class, the gradient is calculated. This means that if I change the input slightly, how much will the prediction change?

$$\text{Video_Relevance} = \text{Video_Input} \odot \text{Gradient_Video}$$

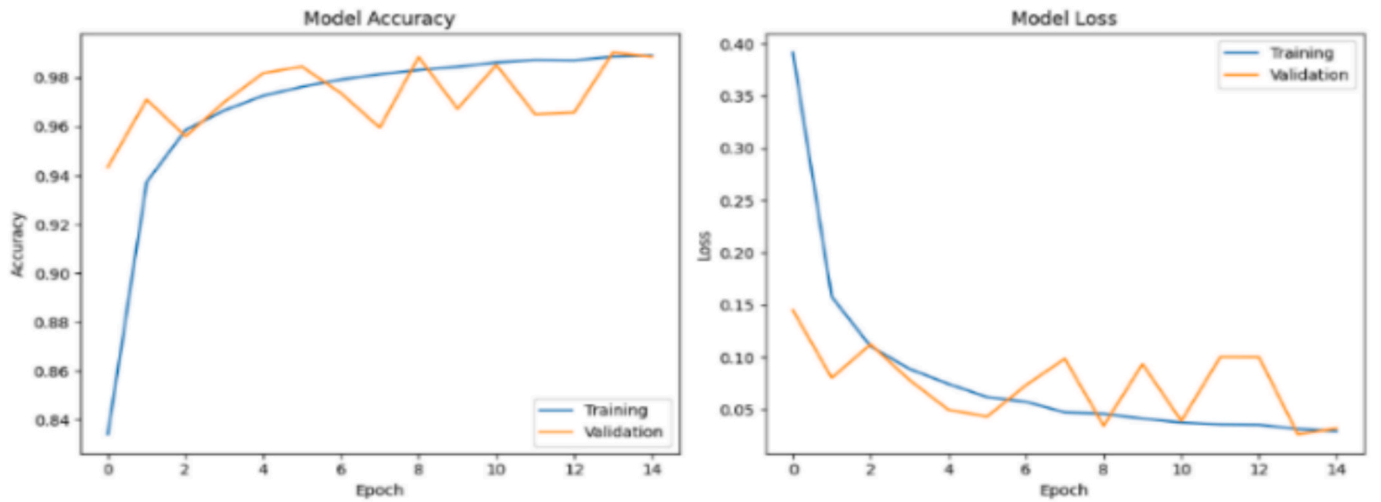


Fig. 6. Values in the training process of the deepfake image model.

Table 1

Accuracy and loss values of the Deepfake image model.

	Train	Validation
Accuracy	98.87%	99.03%
Loss	0.0314	0.0263

$$\text{Audio_Relevance} = \text{Audio_Input} \odot \text{Gradient_Audio}$$

$\odot$ symbol denotes element-wise multiplication, meaning each pixel is multiplied by its own gradient and each audio sample by its own gradient. This formula allows both the feature value and its effect to be

combined.

Finally, the method used is SHAP. MFCC feature extraction was performed for the audio features. Twenty MFCC features are obtained using Librosa. First, a prediction is obtained from the model using the original video and audio, and the model's result is examined without changing anything to see if it is real or fake.

Then, the first MFCC feature is set to 0, and when this is given to the model, it is calculated whether the predicted value has changed and by how much. If it does not change, MFCC feature 1 means that its effect on the predicted value is 0. This process is repeated for all 20 MFCC values, and scores are generated. The SHAP value is the difference between the baseline prediction and the modified prediction. After this is done for each feature, the audio features contributing to the model's prediction

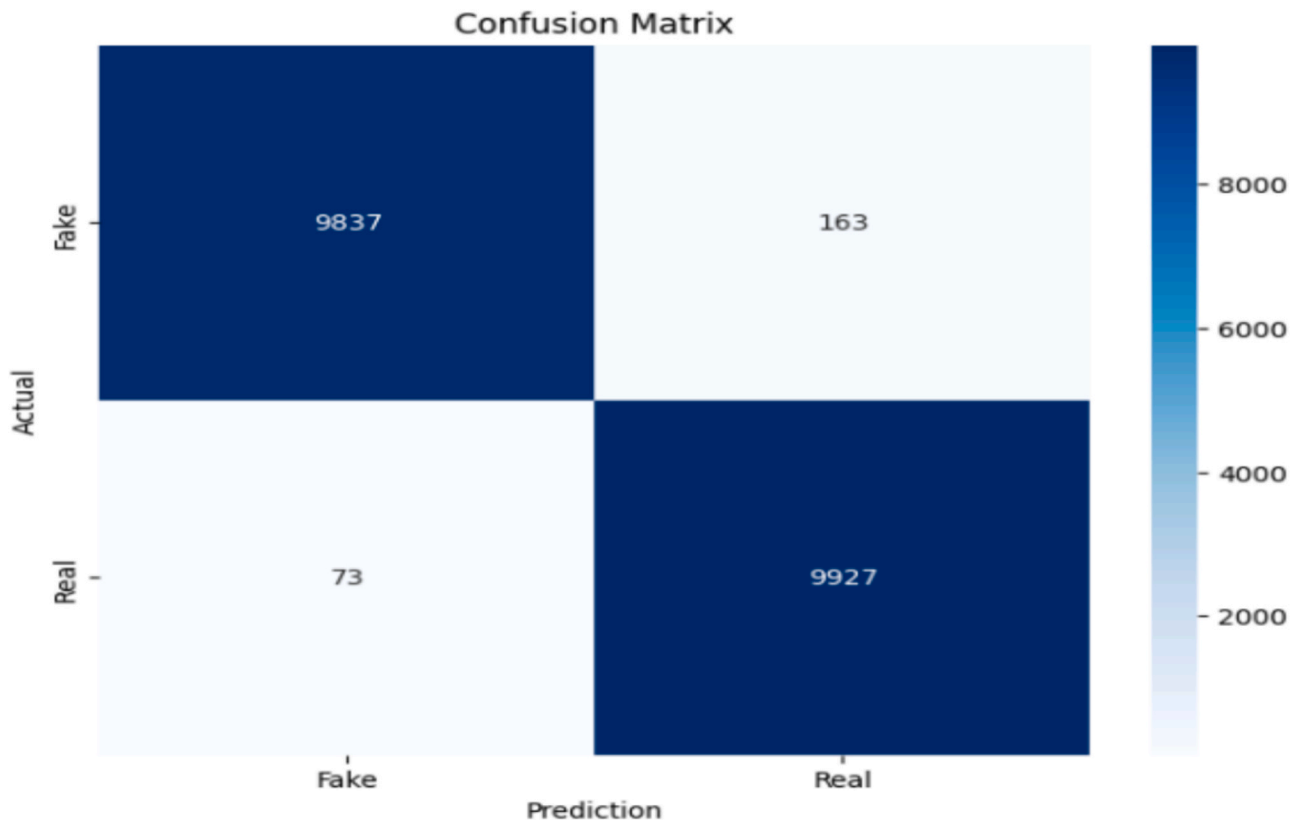


Fig. 7. Confusion matrix of the Deepfake image model.

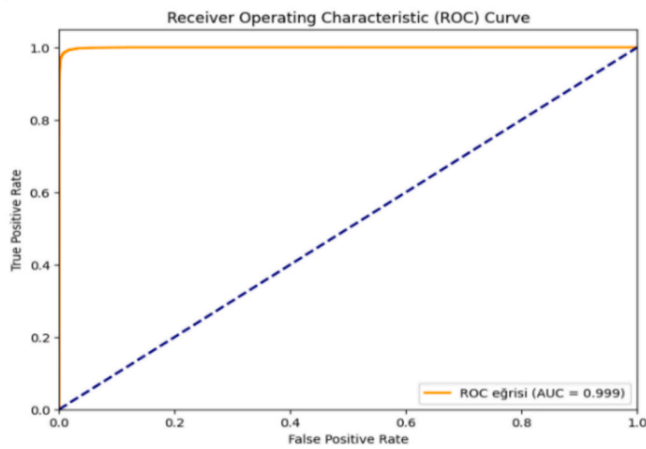


Fig. 8. ROC-AUC values of the Deepfake Image Model.

Table 2

Deepfake image model classification report.

	Precision	Recall	F1-Score	Support
Fake	0.99	0.98	0.99	10,000
Real	0.98	0.99	0.99	10,000
Accuracy			0.99	20,000
Macro avg	0.99	0.99	0.99	20,000
Weighted avg	0.99	0.99	0.99	20,000

are identified, and the model becomes transparent.

4. Experimental results

After the data was trained with the models, each model was measured separately using evaluation parameters. These parameters are accuracy, recall, precision, F1-score, confusion matrix, classification report, and ROC-AUC graph.

4.1. Deepfake image

The model training process for deepfake image detection was completed using the Inception v3 architecture. During the training phase, loss and accuracy values were recorded to monitor the model's learning dynamics. The loss and accuracy graphs in Fig. 6 clearly show the model's development over 15 epochs. The accuracy and loss values for the training and validation data are also included numerically in Table 1.

To evaluate the model's performance more comprehensively, a confusion matrix was created from the classification results of the test data. This matrix provided a detailed view of the number of true positives, true negatives, false positives, and false negatives. Only 236 images were misclassified in the test data, which consisted of 20,000 images. Fig. 7 shows the confusion matrix image identified. In addition, a ROC curve was generated and the area under the curve (ROC-AUC) graph was calculated to analyse the model's classification success in greater depth. This metric contributed to the reliability analysis of the study by demonstrating the model's ability to distinguish between positive and negative examples independently of the threshold value. An accuracy value of 99.9% was achieved. This result is shown in Fig. 8.

Finally, performance metrics such as accuracy, precision, sensitivity, and F1-score are detailed in the classification report. By demonstrating that the model shown in Table 2 exhibits consistent performance at a 99% rate across different metrics, it enables a comprehensive evaluation of the results. Additionally, a loss value of 0.0344 was obtained for the test.

4.2. Deepfake video

The aim was to detect fake videos not only frame by frame but also temporally. For this reason, two models were evaluated: one that trained frame by frame and one that trained temporally. Both models achieved a high accuracy. Fig. 9 shows the accuracy and loss values during the training process for the frame model's training and validation data. The image in Fig. 10 shows that despite the hybrid model having 25 epochs, training stopped early at the 9th epoch, preventing overfitting.

For the frame model, obtained from 400 fake and 400 real videos:
for the real class: 2839 training, 608 validation, 610 test.

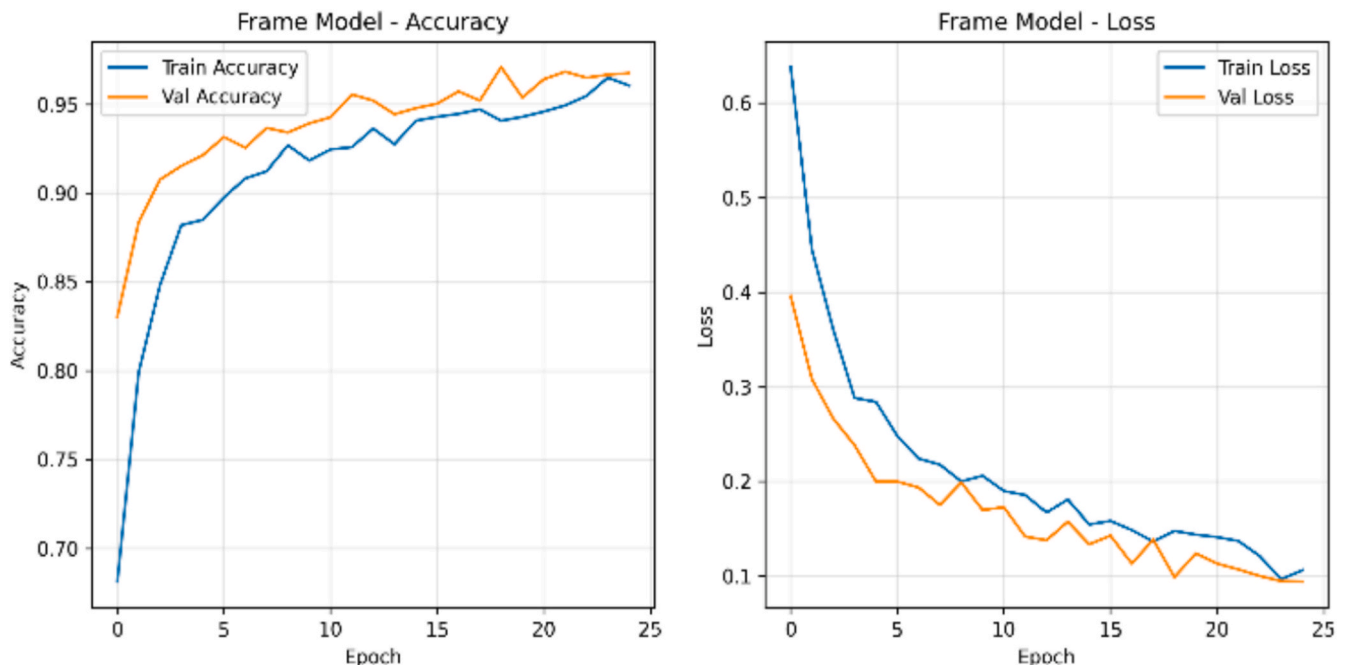


Fig. 9. Values in the training process of the Deepfake video frame model.

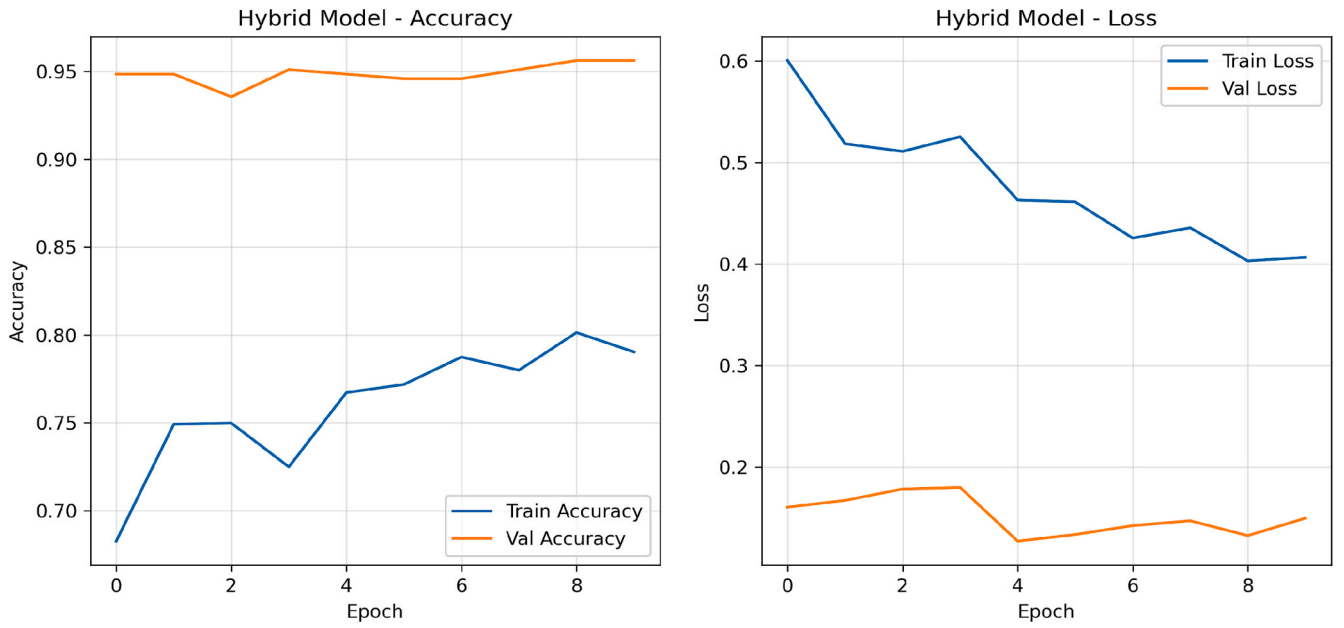


Fig. 10. Values in the training process of the Deepfake video hybrid model.

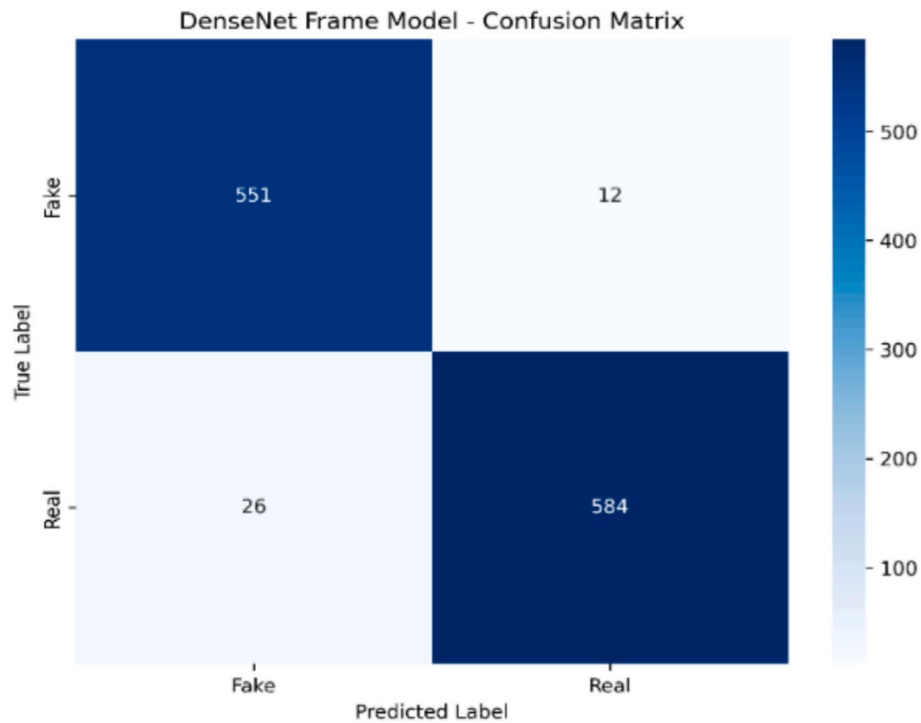


Fig. 11. Confusion matrix of the Deepfake video frame model.

For the fake class: 2623 training, 562 validation, 563 test.

In total, training and testing were performed with 5,462 training frames, 1,170 validation frames, and 1,173 test frames. As a result, Fig. 11 shows the classification predictions obtained by the frame model in the test results. For the hybrid model, 800 videos were divided into 560 training, 120 validation, and 120 test videos. The training sequence is 1726, validation is 388, and the test sequence count is 351. Fig. 12 shows the classification result obtained with the confusion matrix.

The classification report for the frame model shown in Table 3 is provided. An accuracy rate of 96.76% was achieved, while the loss value was 0.098.

The classification report for the hybrid model is shown in Table 4. A loss value of 0.1808 was obtained with an accuracy rate of 90.88%.

The frame model shown in Fig. 13 yields a ROC-AUC value of 99.38%, while the hybrid model shown in Fig. 14 yields a ROC-AUC value of 98.06%, demonstrating high performance.

4.3. Deepfake audio

To detect fake voices, separate model training was conducted comprehensively for both tabular features and time-frequency. The dataset contains a total of 31,779 wav files. Of these, 19,963 are real and

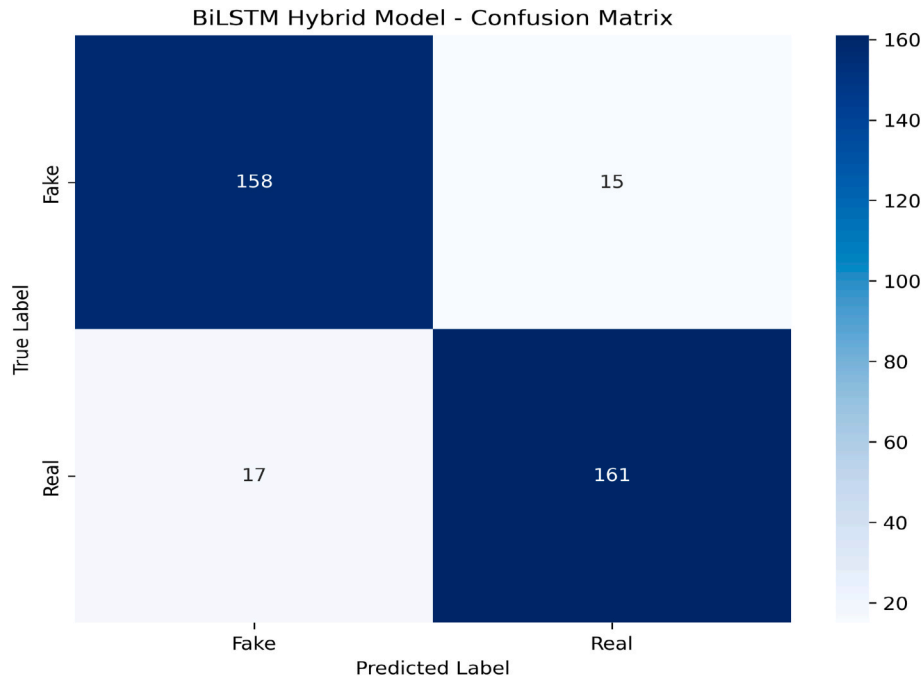


Fig. 12. Confusion matrix of the Deepfake video hybrid model.

Table 3

Classification report of the Deepfake video frame model.

	Precision	Recall	F1-Score	Support
Fake	0.9549	0.9787	0.9667	563
Real	0.9799	0.9574	0.9685	610
Accuracy			0.9676	1173
Macro avg	0.9674	0.9680	0.9676	1173
Weighted avg	0.9679	0.9676	0.9676	1173

Table 4

Classification report of the Deepfake video hybrid model.

	Precision	Recall	F1-Score	Support
Fake	0.9029	0.9133	0.9080	173
Real	0.9148	0.9045	0.9096	178
Accuracy			0.9088	351
Macro avg	0.9088	0.9089	0.9088	351
Weighted avg	0.9089	0.9088	0.9088	351

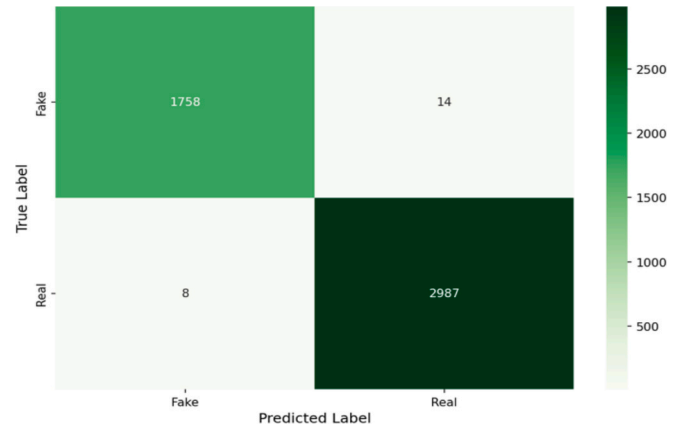


Fig. 14. Confusion Matrix of the Deepfake Audio CNN + BiLSTM Model.

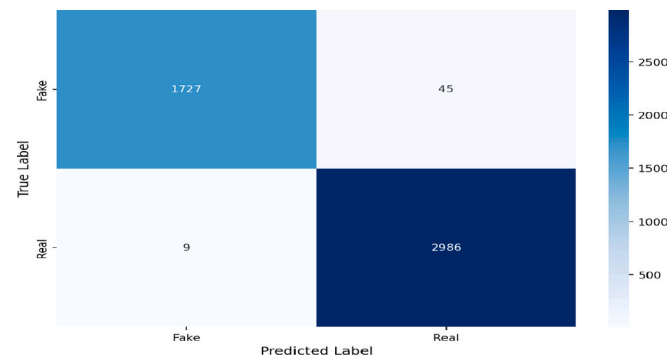


Fig. 13. Confusion matrix of the Deepfake Audio XGBoost Model.

11,816 are fake voice files. The training data was divided into 22,245, validation into 4,767, and testing into 4,767. The classification reports for both models are compared and shown in detail in Table 5. Accuracy

Table 5

Comparison of Accuracy Values for XGBoost and CNN + BiLSTM Models for Deepfake Audio.

	XGBoost (%)	CNN + BiLSTM (%)
Accuracy	98.87	99.54
Precision	98.52	99.53
Recall	99.70	99.73
F1-Score	99.10	99.63

above 98% were achieved for both tabular features and temporal features.

The confusion matrix visualisation provided in Fig. 13 belongs to the XGBoost model, which misclassified only 54 audio files in the test data. In Fig. 14, the CNN + BiLSTM model misclassified 22 audio files. Consequently, both models achieve high accuracy.

Fifty epochs were selected for the CNN + BiLSTM model. However, training stopped at the 33th epoch due to the parameters added to prevent overfitting. The accuracy and loss values of the training and validation data were recorded throughout the training period and are

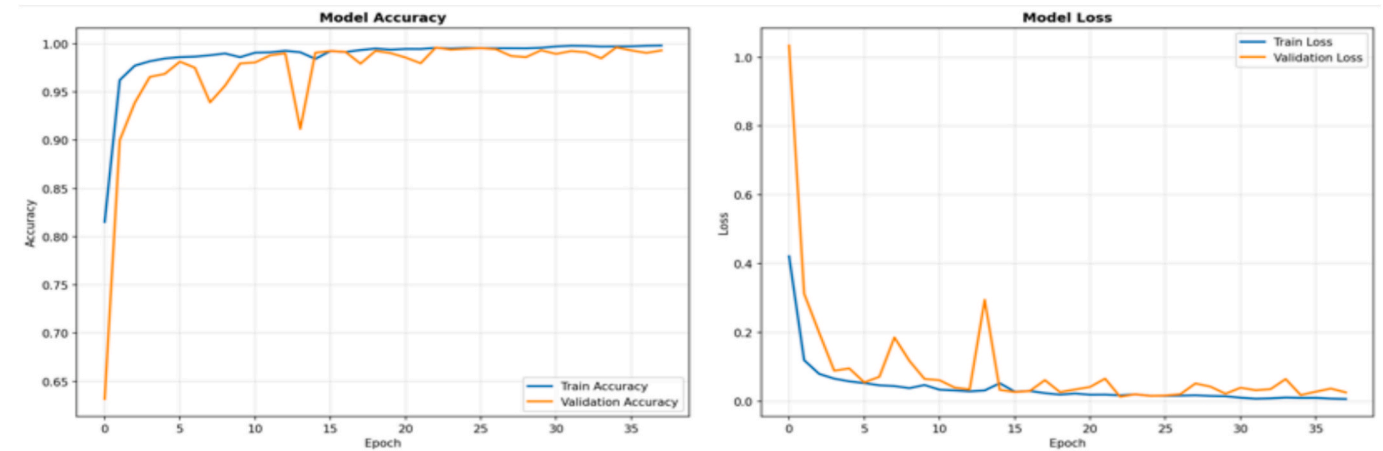


Fig. 15. Values in the training process of the Deepfake Audio CNN + BiLSTM Model.

Table 6
Accuracy and Loss Values of the Deepfake Audio CNN + BiLSTM Model.

	Training	Validation
Accuracy	99.72%	99.62%
Loss	0.0104	0.0186

shown in Fig. 15. The values of the recorded model for the training and validation data are added to Table 6.

Additionally, in addition to these evaluation parameters, the ROC-AUC graphs of both models shown in Fig. 16 have been compared. XGBoost achieved an accuracy value of 99.95%, while the CNN +

BiLSTM model achieved 99.98%.

4.4. Deepfake multimodal (Audio + Visual)

The evaluation parameters obtained as a result of training the multimodal model have been recorded. A total of 6,000 videos were collected in the test data. Fig. 17 details the model’s correct and incorrect classifications using a confusion matrix.

When evaluated using the assessment parameters obtained from the test dataset in Table 7, the hybrid model consisting of DenseNet169 + BiLSTM + Cross Attention achieved a high accuracy. Table 8 records the accuracy and loss values obtained from the training and validation data for the epoch step that yielded the best result during model training.

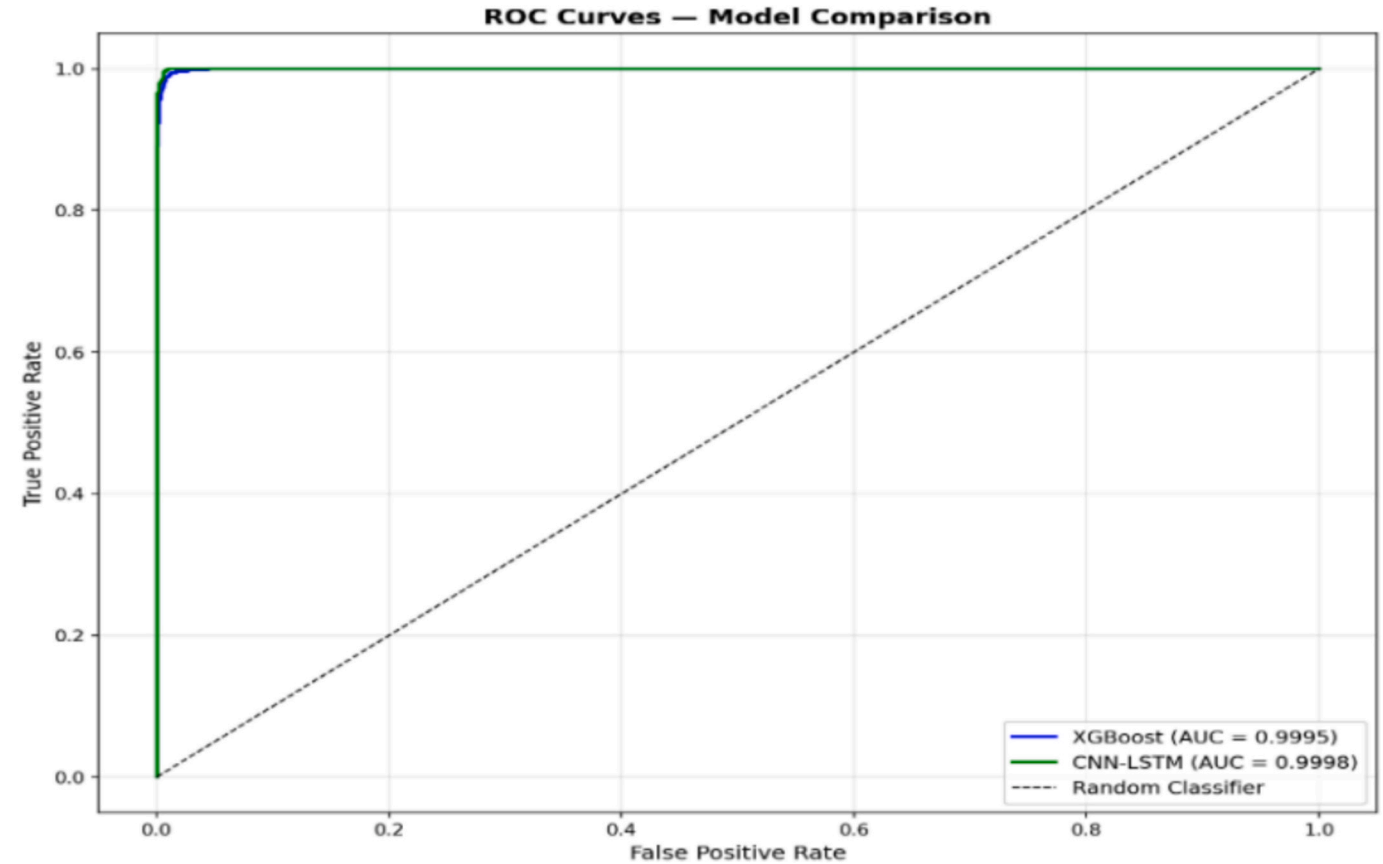


Fig. 16. ROC-AUC Graph for XGBoost / CNN + BiLSTM Models for Deepfake Audio.

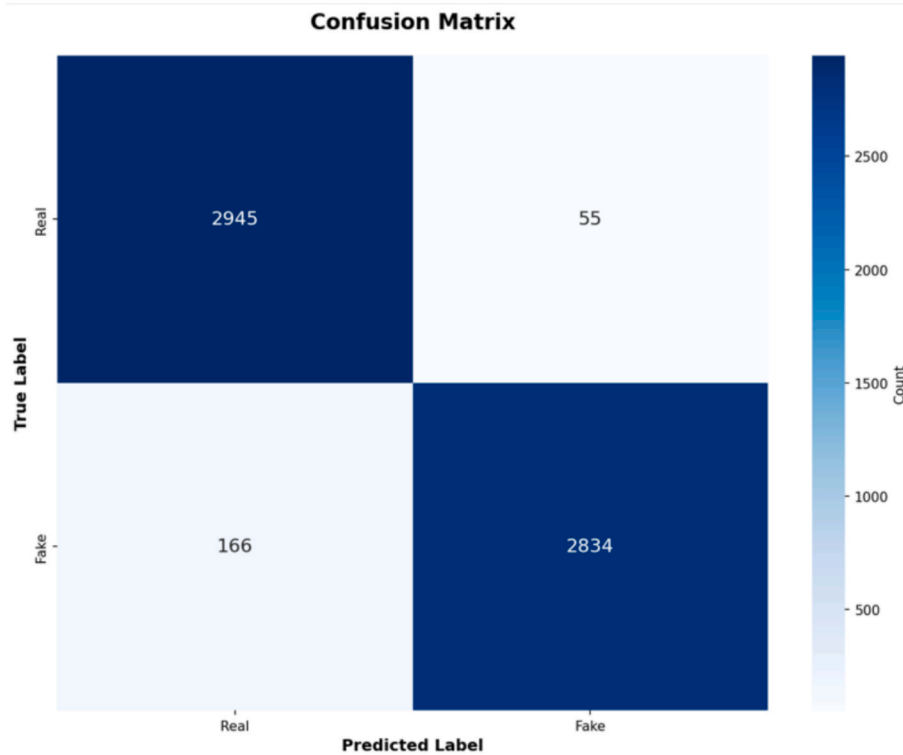


Fig. 17. Confusion Matrix of the Hybrid Model for Deepfake Multimodal.

Table 7

Accuracy Values of the Deepfake Multimodal Hybrid Model.

Accuracy	Precision	Recall	F1-Score	Specificity
96.32%	98.10%	94.47%	96.25%	98.17%

Table 8

Accuracy and Loss Values of the Deepfake Multimodal Hybrid Model.

	Training	Validation
Accuracy	98.94%	94.07%
Loss	0.0307	0.1870

The model achieved an accuracy value of 99.68% in the ROC-AUC graph shown in Fig. 18, which is another evaluation parameter.

4.5. Ablation study: effect of cross-modal attention

To experimentally evaluate the contribution of the cross-modal attention mechanism to multimodal detection performance, an ablation study was conducted. In this multimodal model, only the cross-attention module was removed and replaced with a simple concatenation-based fusion. Both model variants were trained on the LAV-DF dataset using the same hyperparameters, data splits, and class-balanced sampling. The results of the ablation study are presented in Table 9.

The results in Table 9 demonstrate that the cross-modal attention mechanism provides a significant performance boost. The proposed model achieved 1.85% higher accuracy and 1.81% higher F1-score compared to the ablation variant. Additionally, the increases in precision and specificity, 3.21% and 3.24% respectively, indicate that the cross-attention module is particularly effective in reducing false positives. The observed 0.91% improvement in AUC further confirms an overall enhancement in the model's discriminative ability. These

findings clearly support that the cross-modal attention mechanism is a performance-enhancing component in multimodal deepfake detection.

4.6. Statistical reliability analysis

To comprehensively evaluate the robustness and generalizability of the proposed framework, advanced statistical analyses were conducted across all modalities. These analyses include convergence stability, bootstrap-based confidence interval estimation, Wilcoxon signed-rank test, and Friedman test, providing in-depth quantitative evidence of model performance beyond standard accuracy metrics.

For the image modality (InceptionV3), 10-fold cross-validation analysis demonstrated an average accuracy of 98.75% and a coefficient of variation (CV) of 0.021, indicating a highly stable learning behavior. Bootstrap resampling ($n = 30$) yielded a 95% confidence interval $[97.73\%, 99.77\%]$ and an average value of $98.72 \pm 0.57\%$, confirming the high reliability and reproducibility of the model's performance. The Wilcoxon signed-rank test ($p = 0.75$) verified that there was no statistically significant deviation in the performance distribution. In the XAI reliability (faithfulness) analysis, insertion AUC = 0.7337 ± 0.1590 and deletion AUC = 0.4860 ± 0.1822 with a positive difference of 0.2477 indicated that LIME and LRP-based explanations effectively highlighted the regions influencing model decisions. The faithfulness correlation of 0.4623 ± 0.1278 showed a moderate alignment between explanations and model behavior.

For the video modality, the DenseNet169 frame-based model achieved $98.75 \pm 1.91\%$ average accuracy (CV = 0.0193), demonstrating high convergence stability. Bootstrap analysis produced a 95% confidence interval $[97.27\%, 99.70\%]$, supporting the statistical reliability of this performance. The Wilcoxon test comparing the frame model and BiLSTM-based hybrid model ($p = 0.0020$) revealed that the frame-level approach (98.75%) significantly outperformed the sequence-based model (75.65%), indicating that spatial features in video data are more dominant than temporal modeling. The Friedman test ($\chi^2 = 20.0$, $p < 0.001$) confirmed significant performance differences across different model configurations. Grad-CAM-based XAI analysis showed

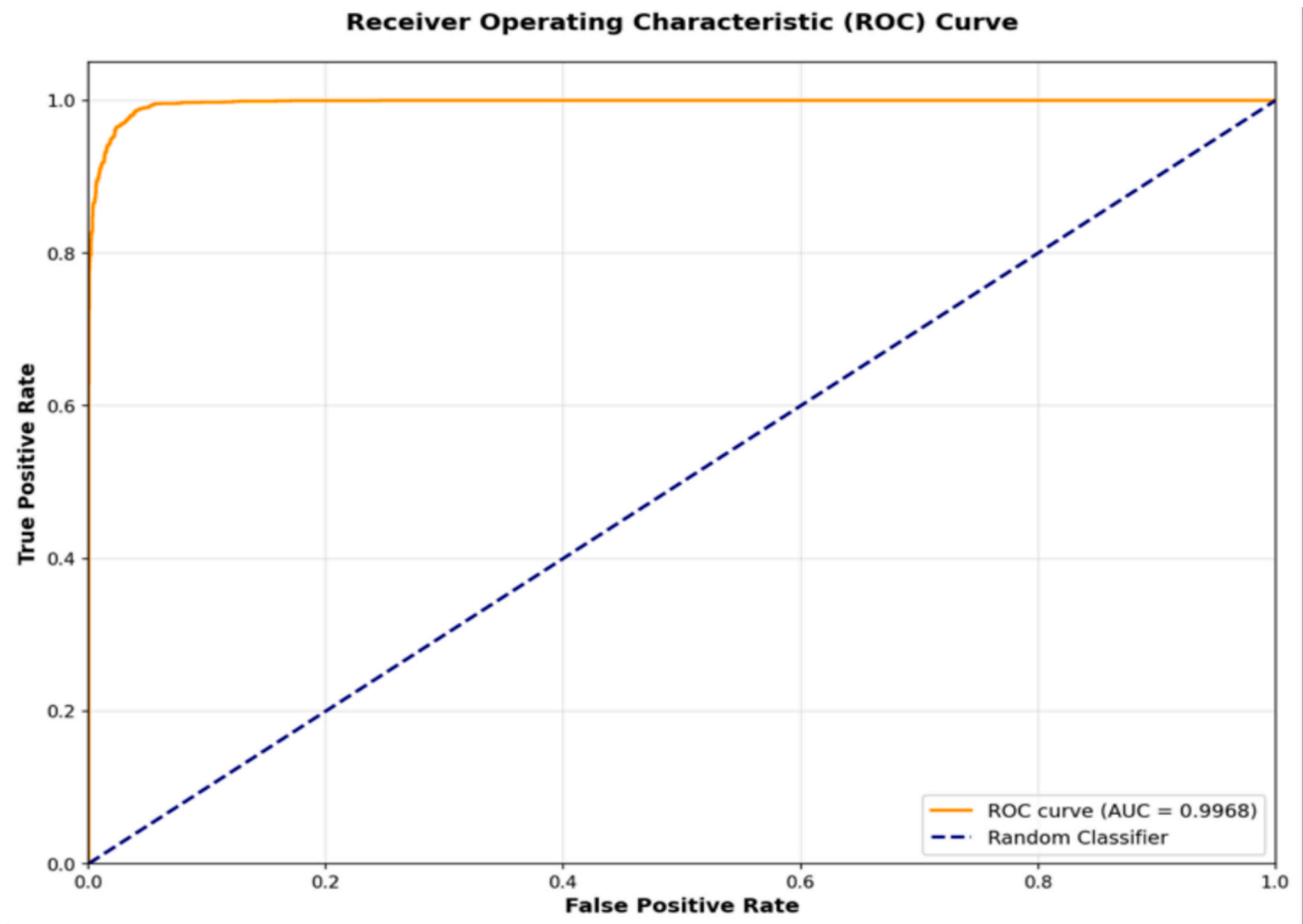


Fig. 18. ROC-AUC Graph of the Deepfake Multimodal Hybrid Model.

Table 9

Impact of cross-modal attention on multimodal Deepfake detection performance.

Configuration	Accuracy	Precision	Recall	F1-Score	Specificity	AUC
With Cross-Attention Multimodal (Proposed)	96.32%	98.10%	94.47%	96.25%	98.17%	99.68%
Without Cross-Attention (Ablation)	94.47%	94.89%	94.00%	94.44%	94.93%	98.77%
Δ (Proposed – Ablation)	+1.85%	+3.21%	+0.47%	+1.81%	+3.24%	0.91%

insertion = 0.7524 ± 0.0529 and deletion = 0.5914 ± 0.0246 , with a faithfulness correlation of 0.2409 ± 0.2408 , suggesting that explanations are locally consistent but may vary across samples.

The CNN-LSTM-based audio model, trained on Mel-spectrograms, achieved 99.50% average accuracy with CV = 0.0151, indicating high convergence stability. Bootstrap resampling ($n = 30$) provided a 95% confidence interval [98.50%, 100.00%], confirming statistical reliability. Wilcoxon ($p = 0.5000$) and Friedman ($p = 0.1353$) tests

comparing XGBoost and CNN-LSTM models showed no significant differences, indicating that the performances of both models are statistically equivalent. In the XAI reliability analysis, insertion AUC = 0.6754 ± 0.2548 , deletion AUC = 0.3379 ± 0.2102 , and Ins-Del difference = 0.3375 were measured. Faithfulness correlation was 0.5507 ± 0.5908 , showing that explanations generally reflect model decisions but vary significantly across samples. The relatively lower AUC values are due to the high heterogeneity of temporal and frequency components in audio

Table 10

Statistical Reliability Summary Across All Modalities.

Modality / Model	Avg Accuracy (%)	CV	Bootstrap Mean	Wilcoxon p-value	Friedman p-value	XAI Insertion AUC	XAI Deletion AUC	XAI Faithfulness Corr.
Image (InceptionV3)	98.72 ± 0.57	0.021	[97.73, 99.77]	0.75	–	0.7337 ± 0.1590	0.4860 ± 0.1822	0.4623 ± 0.1278
Video (DenseNet169 Frame)	98.75 ± 1.91	0.0193	[97.27, 99.70]	Frame vs BiLSTM: 0.0020	$\chi^2 = 20.0, p < 0.001$	0.7524 ± 0.0529	0.5914 ± 0.0246	0.2408 ± 0.2408
Audio (CNN-LSTM)	99.50	0.0151	[98.50, 100.00]	XGBoost vs CNN-LSTM: 0.5000	$p = 0.1353$	0.6754 ± 0.2548	0.3379 ± 0.2102	0.5507 ± 0.5908
Multimodal (Audio + Visual)	90.17 ± 2.68	0.099	[85.72, 94.82]	Ses vs Video: 0.0156	$\chi^2 = 11.31, p = 0.0035$	0.6207 ± 0.234	0.3102 ± 0.171	0.5475 ± 0.1851

Table 11

Performance Comparison of Different Backbone Architectures in the Multimodal Model.

Model	Accuracy	Precision	Recall	F1-Score	Specificity	AUC
MobileNetV2 + BiLSTM + CNN(Audio)	77.20%	84.43%	66.70%	74.53%	87.70%	86.54%
ResNet50V2+ BiLSTM + CNN(Audio)	81.85%	90.85%	70.83%	79.60%	92.87%	93.44%
Ours	96.32%	98.10%	94.47%	96.25%	98.17%	99.68%

signals, causing XAI methods to exhibit more variable performance on audio than on visual data.

For the multimodal model, bootstrap analysis ($n = 30$) showed $90.17 \pm 2.68\%$ average accuracy with a 95% confidence interval [85.72%, 94.82%], indicating an overall stable model structure. The Wilcoxon test comparing audio and video channels ($p = 0.0156$) confirmed statistically significant differences in contributions, with the video channel (87.00%) being more dominant than the audio channel (71.00%). This suggests that the dataset is more discriminative in terms of visual features, with audio playing a complementary role in the multimodal scenario. The Friedman test ($\chi^2 = 11.31$, $p = 0.0035$) demonstrated that the multimodal fusion strategy provides significantly higher information content than individual modalities. In terms of XAI reliability, video frame saliency analysis showed insertion AUC = 0.6207 ± 0.234 , deletion AUC = 0.3102 ± 0.171 , and Ins-Del difference = 0.3105. Faithfulness correlation was 0.5475 ± 0.1851 , indicating moderate alignment between explanations and decision-making. This highlights that Integrated Gradients and Grad-CAM-based methods are useful for visually interpreting model decisions.

Overall, the findings demonstrate that the proposed framework exhibits strong performance in terms of high accuracy, statistical reliability, and explainability, particularly in video and multimodal scenarios. The results of the statistical evaluation are summarized in Table 10.

Table Explanations:

- CV (Coefficient of Variation): Indicates the convergence stability of the model. Lower values correspond to higher stability.
- Wilcoxon Test: Tests the statistical differences between model configurations.
- Friedman Test: Measures significant performance differences across different model configurations.
- XAI Metrics (Insertion / Deletion / Faithfulness): Show the consistency and effectiveness of model explanations with respect to the decision-making process.

4.7. Additional evaluation with alternative models

To further reinforce the effectiveness of the proposed approach, additional evaluations were conducted using different backbone architectures in the multimodal setting. As presented in Table 11, while MobileNetV2- and ResNet50V2-based models demonstrated competitive

performance, the proposed framework consistently outperformed them across all evaluation metrics. Notably, substantial improvements were observed in accuracy, recall, and F1-Score.

Alternative models were also tested for other modalities; however, only the most representative and comparable results are presented. This ensures that the primary focus of the study remains on cross-modal interaction modeling and explainability. Table 11 clearly illustrates the superior performance of the proposed multimodal framework when compared against various backbone architectures.

4.8. Explainable artificial intelligence and Gradio interface

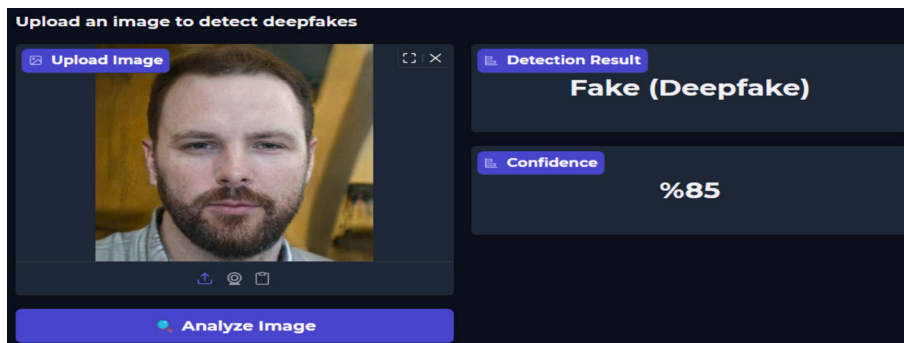
When training models for each media type, the model step achieving the highest accuracy value was recorded. These recorded models were then reloaded alongside explainable artificial intelligence methods and integrated into the analysis process. In the final stage, a Gradio interface was created to enable all models to be used on a single platform, and a separate section was designed for each media type so that both the detection process and the relevant explainability outputs could be presented to the user.

4.8.1. Deepfake image

An image is uploaded to the section of the system allocated for images, after which the model decides whether it is real or fake and displays the percentage it has detected. Fig. 19 shows the class of the uploaded image. Then, as shown in Fig. 20, the areas affected by the model's decision that the uploaded original image is fake are shown using saliency and lime, making it clear which areas are fake. Furthermore, as shown in Fig. 21, the textual explanation shows that Saliency focuses on the cheeks, eye area, and nose, while LIME focuses on the chin, mouth area, and cheeks.

4.8.2. Deepfake video

The video has been loaded into the Gradio system using both the frame and hybrid models. A video is loaded into the designated video section of the Gradio interface. The video is analysed first using the frame model and then the hybrid model to determine whether it is real or fake. Each second is taken as one frame. Grad-CAM is then applied to each frame to show the ratios of fake and real. As a result, it is possible to determine in which second manipulation has occurred. If there is falsification in even one frame, the result is false. Fig. 22 shows the result of the uploaded video. Figs. 23 and 24 show the Grad-CAM analysis of each

**Fig. 19.** Gradio – Deepfake image detection.

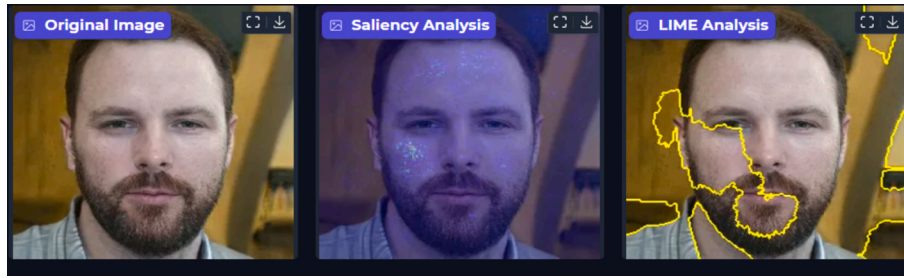


Fig. 20. Gradio – Deepfake Image XAI Images.

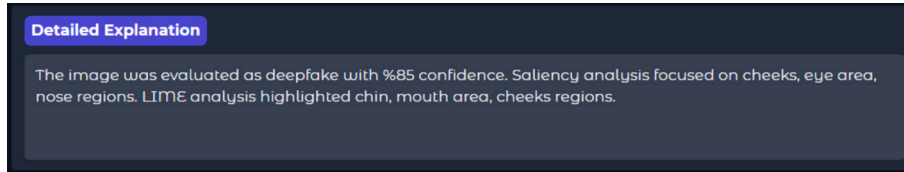


Fig. 21. Gradio – Deepfake image textual description.

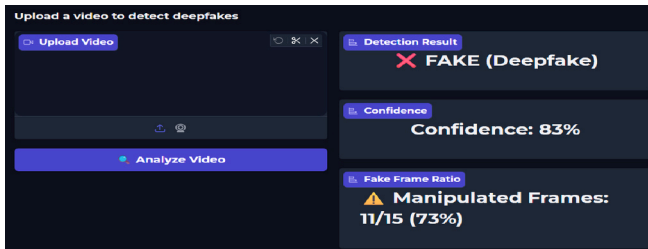


Fig. 22. Gradio – Deepfake video detection.



Fig. 23. Gradio – Deepfake Video Grad-CAM Images.



Fig. 24. Gradio – Deepfake Video Grad-CAM Images.

frame's results. In this example, it is understood that 11 out of 15 frames of the fake video have been manipulated. The red colours indicate the areas that most influence the model's decision. Blue indicates less influence.

4.8.3. Deepfake audio

After loading the model for sound, any file with a sound extension can be loaded into the system. It is analysed as shown in Fig. 25. Then, as shown in Fig. 26, both models make predictions and the results are evaluated. There are four models in the explanation section. First, both local and global features are compared using SHAP and LIME. Fig. 27 shows the LIME analysis. Fig. 28 shows the beeswarm model of the SHAP model. In this way, by comparing the two models, it can be seen how much each feature influences the model's decision.

Fig. 27 shows the 10 features with the greatest impact using LIME. Blue indicates that the feature affects the fake class. Orange indicates that the feature positively affects the real class. The Spectral_Stat_9 feature has the greatest impact on the fake class. The MFCC_25 value also positively affects the true class.

However, some values, such as MFCC_19, have a negative value but a positive effect. This means that the model sees it as a feature specific to the true class and pushes the prediction away from the fake class towards the true class. This shows that the model is learning in which direction and in which range a feature distinguishes the class.

In Fig. 28, all features have been evaluated using SHAP. As with LIME, the feature with the most significant negative impact is Spectral_Stat_9. In SHAP, the MFCC_25 feature has a high negative impact. MFCC_15 has a high positive impact. In this way, all features can be evaluated, and the model's decision becomes understandable.

Figs. 29 and 30 show the temporal evaluation of the LRP model on the spectrogram using Grad-CAM. The Grad-CAM model indicates which regions it focuses on. It concentrates on the 20–60 mel band and the 20–40,100–120 time segment. The LRP model distributes positive/negative contribution values to each input. Its interpretability is slightly more complex but more comprehensive. Areas with high relevance values are shown in yellow. Black areas indicate areas that do not influence the model's decision. In the 60–100 mel range and more detailed time intervals, the model's decision can be interpreted based on the pixels influencing the decision.

4.8.4. Deepfake multimodal

As with other models, a dedicated space has been created for multimodal analysis in the Gradio interface. The user uploads a video, and the model determines whether it is real or fake. As shown in Fig. 31, a video has been uploaded and identified as fake. At the same time, one of the XAI methods is selected prior to the analysis. To enhance interpretability, a description of the selected XAI method is provided, including what it is and how it should be interpreted by the user.

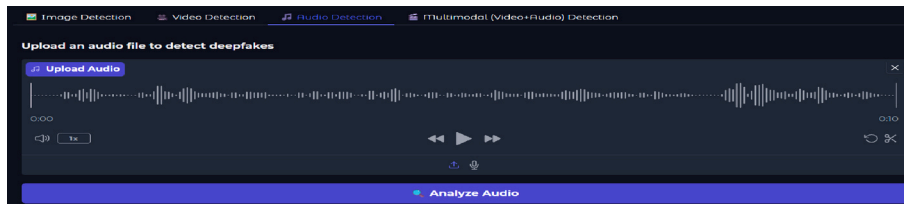


Fig. 25. Gradio – Deepfake audio detection.

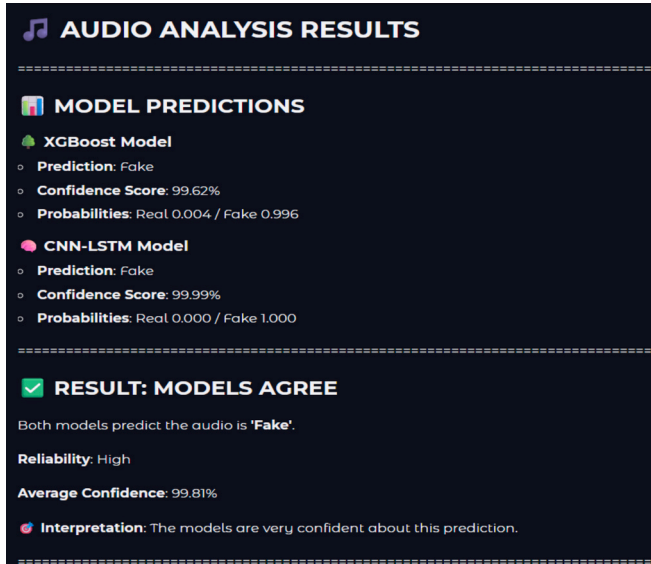


Fig. 26. Gradio – Deepfake audio detection.

The values from the metadata.json structure of the tested video are presented in Fig. 32. It is shown that the video is fake, while the audio has not been altered. Additionally, the manipulation is detected within the time interval of 3.4–3.68 s.

Grad-CAM, one of the explanation methods shown in Fig. 33, has been selected. This provides explainability only for images. Four frames are displayed, and the areas of focus in the face region are shown as a heatmap, making the model's decision understandable.

Fig. 34 shows the Integrated Gradients explanations. Initially, 4 frames are displayed, and the values of the 8 frames' contributions to the model are shown in more detail. This allows for a more detailed examination of the face's movements. Additionally, the Attribution score is calculated for the 8 frames. Looking at these scores, it is seen that the focus is on the first frame, and then manipulation occurs in the 7th and 2th frames. The Amplitude value within the sound and its average are shown in a more smoothed manner. It is understood that the model detects inconsistencies between 3.5 and 4.0.

Fig. 35 shows which pixels the model focuses on for LRP images. Additionally, amplitude values such as Integrated Gradients are examined, thereby providing double verification. Inconsistencies between 3.5 and 4.0 have also been identified here. Finally, the mel spectrogram has been checked temporally.

Finally, the characteristics of the sound are evaluated in Fig. 36. The 15 most influential characteristics are shown. It is understood that the MFCC_11 characteristic of the uploaded video has a greater impact on the model.

As a result of the four explainable models, images were evaluated based on pixels and gradients, while sound could be interpreted both as a feature and temporally.

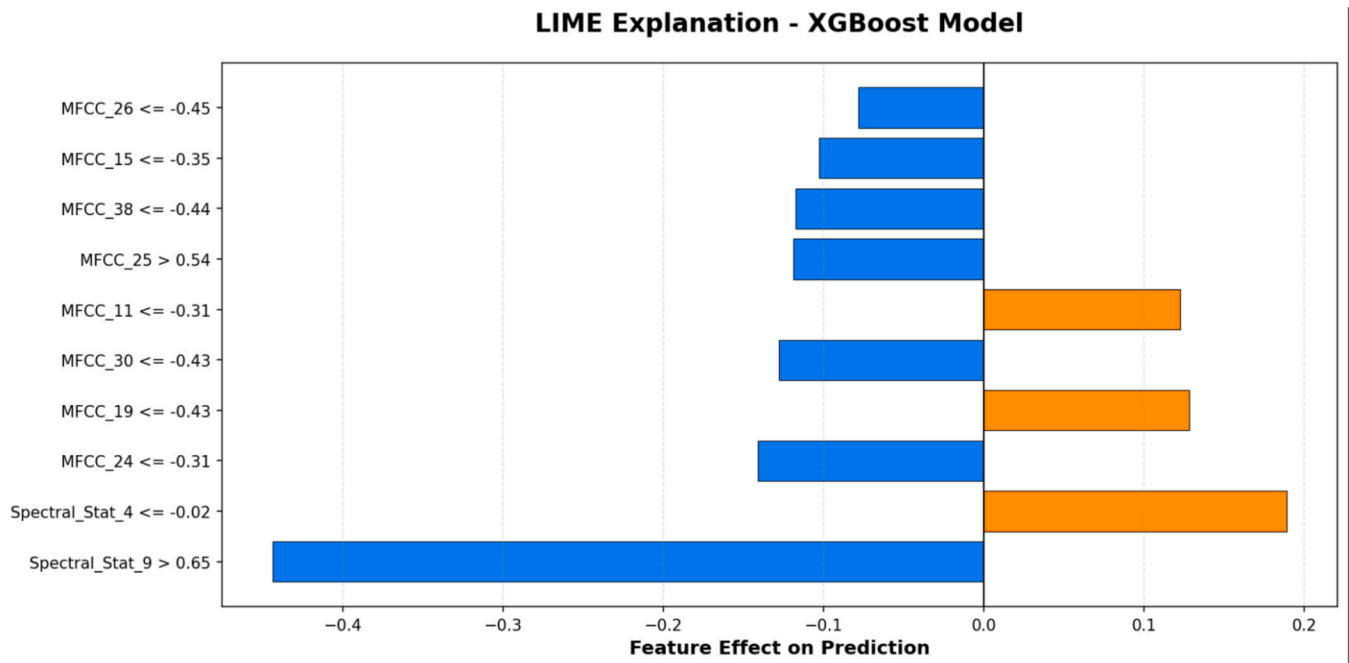


Fig. 27. Gradio – Deepfake audio LIME Table.

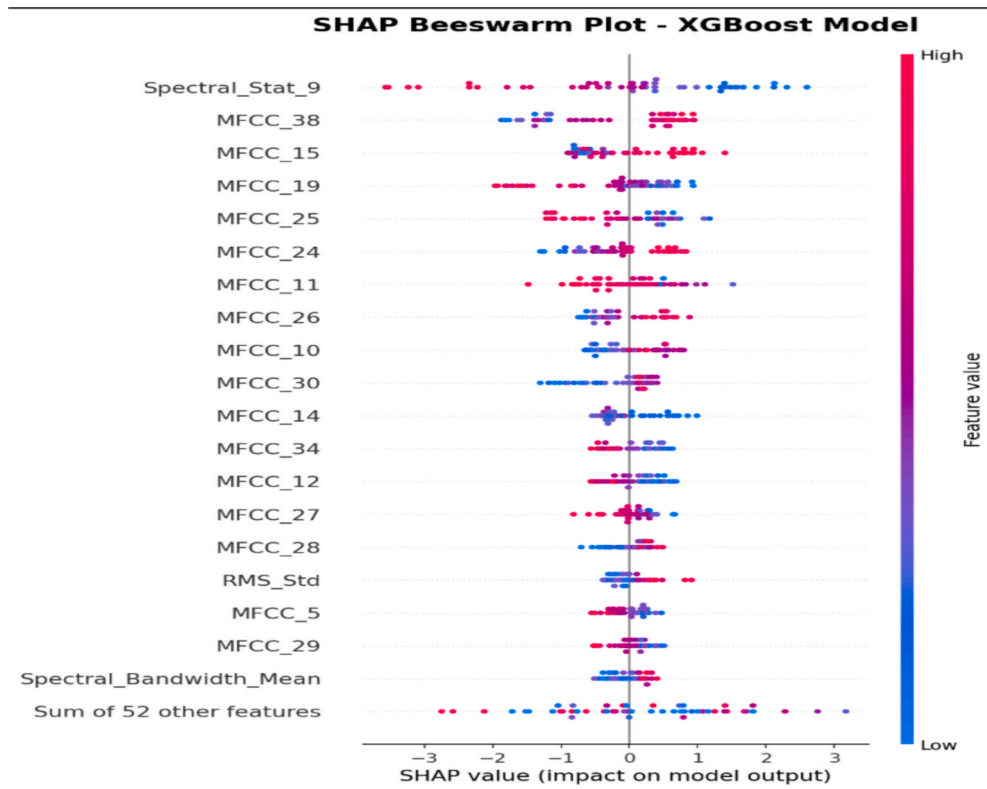


Fig. 28. Gradio – Deepfake audio SHAP Values.

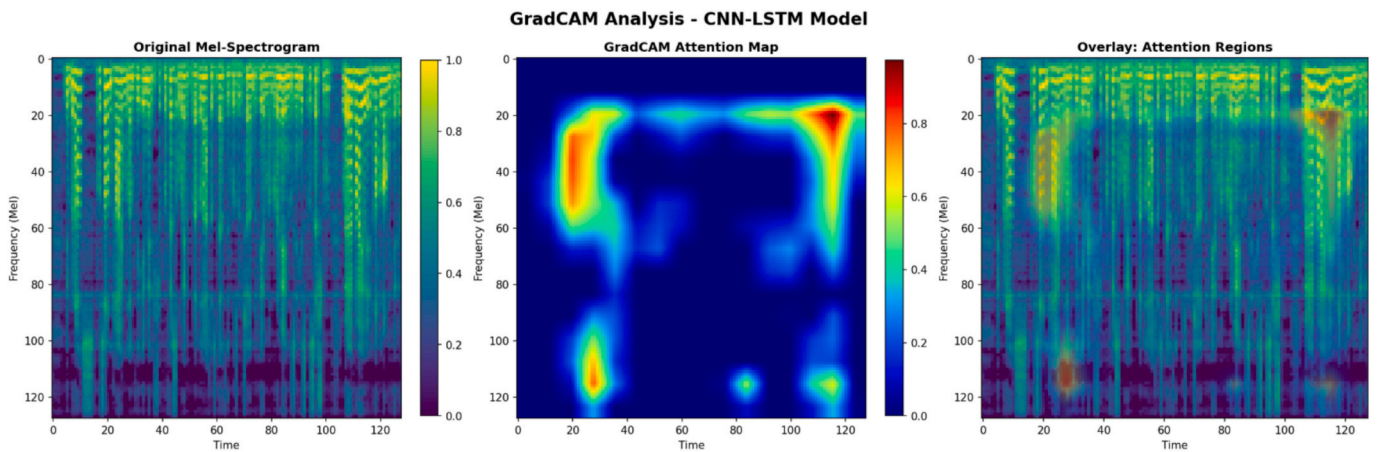


Fig. 29. Gradio – Deepfake Audio Grad-CAM.

4.9. Comparison with deepfake detection methods

Table 12 summarises present studies. Our study analyses all media types simultaneously rather than just one, and includes a system that covers their explainability. Both the internal and external structures of the models can be interpreted in the images. Videos are also analysed not only frame but also temporally, ensuring that each frame can be explained individually. Sound is also analysed and interpreted not only based on features but also temporally, and finally, a video is analysed both visually and aurally, enabling its features to be interpreted in detail. This study stands out from other studies conducted with these features.

4.10. Limitations

Despite strong performance in image, audio, video, and multimodal deepfake detection tasks, some limitations remain. Modality-specific components are optimized through separate training pipelines, which may limit the full exploitation of cross-modal relationships in high-uncertainty or adversarial manipulation scenarios. XAI evaluations, although supplemented with quantitative metrics, show variability in explanation reliability across audio and multimodal modalities. The current framework focuses on binary classification, and while dedicated adversarial attacks have not been applied in the current study, potential attack vectors are discussed, and the development of adversarial training pipelines is planned as future work. These areas remain open directions for further research. Furthermore, we conducted a cross-dataset generalization analysis by evaluating the multimodal model

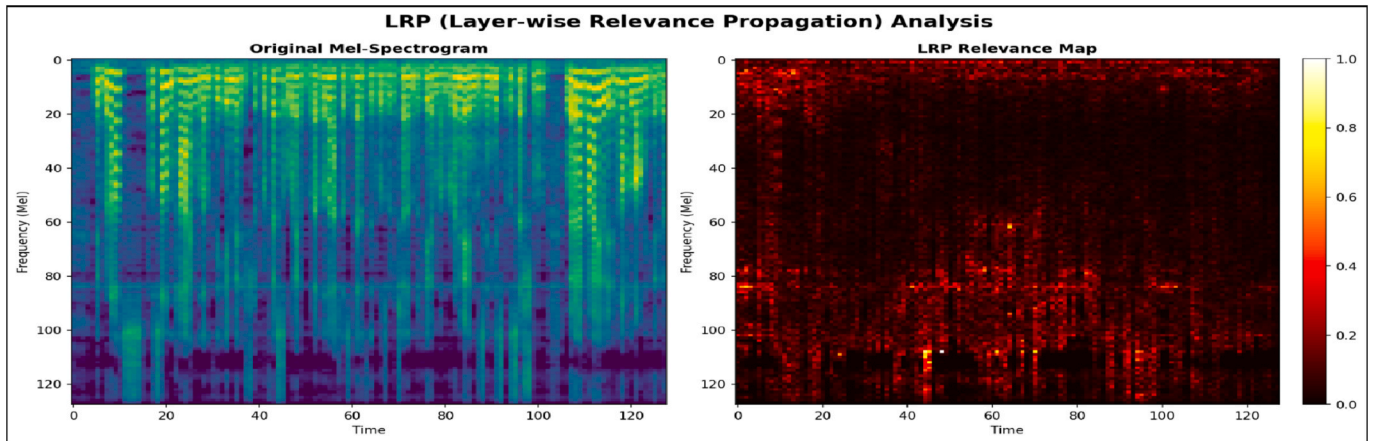


Fig. 30. Gradio – Deepfake Audio LRP.

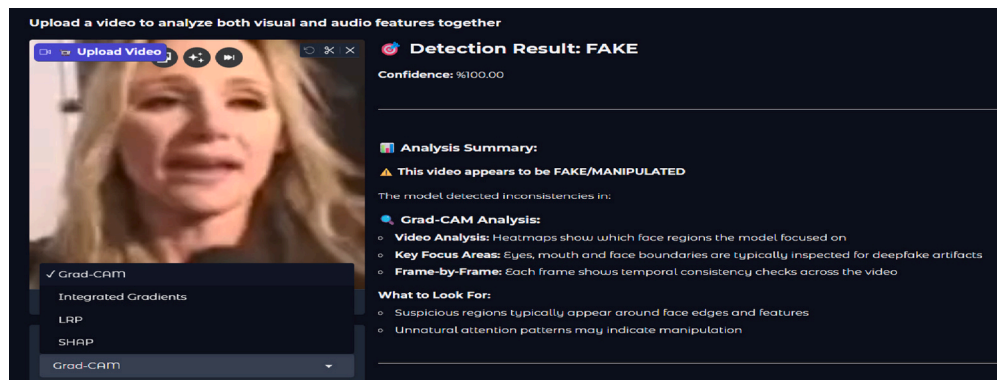


Fig. 31. Gradio – Multimodal Deepfake Detection.

```

446
file "test/000458.mp4"
n_fakes 1
fake_periods [] 1 item
0 [] 2 items
0 3.4
1 3.68
timestamps [] 16 items
duration 5.504
transcript "I Need Your Love attention energy is extremely insignificant would have caused it is or whatever"
original "test/000456.mp4"
modify_video true
modify_audio false
split "test"
video_frames 134
audio_channels 1
audio_frames 86016

```

Fig. 32. Gradio – Multimodal Metadata.json of the Tested Video.

trained on LAV-DF against the FakeAVCeleb v1.2 dataset. The model achieved an AUC of 0.565 under default thresholds, which increased to 0.64 after threshold calibration. This performance difference is expected due to fundamental distributional distinctions between the datasets: LAV-DF primarily involves content-driven lip-sync manipulation with audio artifacts, whereas FakeAVCeleb v1.2 focuses on face-swap forgeries with studio-quality video and largely unaltered audio. Category-level analysis indicated that FakeVideo-FakeAudio samples were most readily detected (65.2%), while RealVideo-FakeAudio and FakeVideo-

RealAudio were more challenging, confirming that the model's discriminative features are most effective when both modalities contain manipulation artifacts. These results align with prior findings in deepfake detection literature, highlighting cross-dataset generalization as an important research direction and motivating future work on domain-adaptive training strategies.

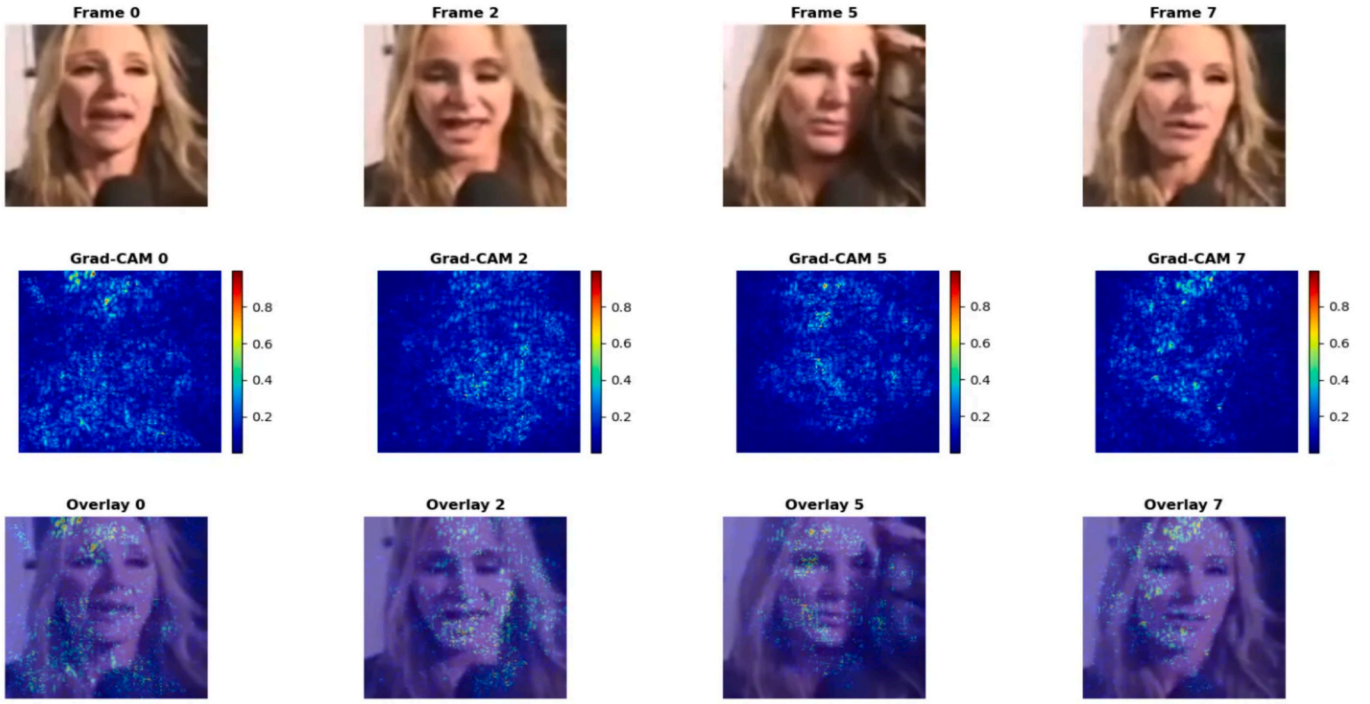


Fig. 33. Gradio – Multimodal Deepfake Grad-CAM.

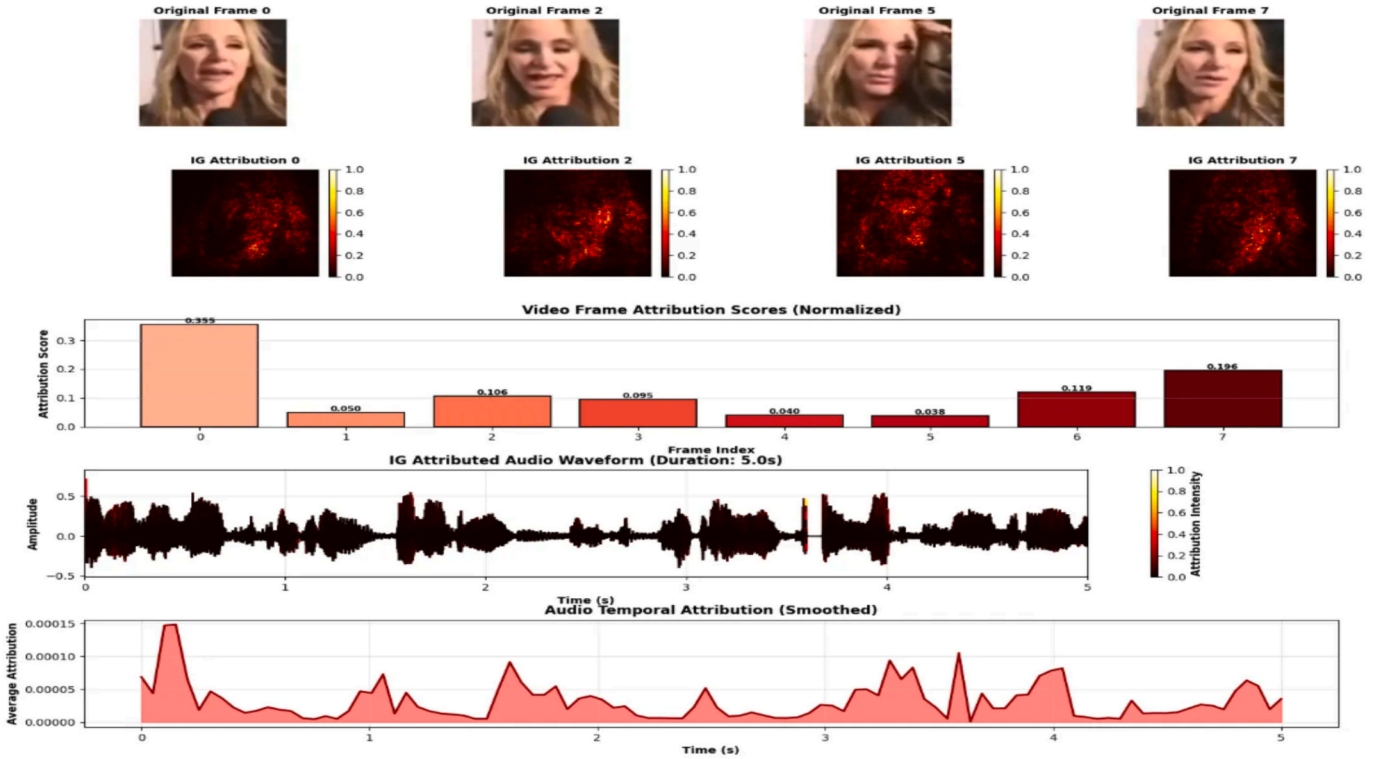


Fig. 34. Gradio – Multimodal Deepfake integrated gradients.

4.11. Discussion

The proposed framework demonstrates strong and statistically validated performance across all four modalities. The high image detection accuracy (99%) reflects InceptionV3's capability to capture fine-grained spatial artifacts generated by StyleGAN-based synthesis, a finding consistently supported by Saliency and LIME explanations focusing on

periocular regions and skin texture boundaries. In video detection, the performance difference between the frame-level DenseNet169 model (96.76%) and the hybrid BiLSTM model (90.88%) was statistically confirmed by the Wilcoxon signed-rank test ($p = 0.0020$), indicating that spatial artifacts are more discriminative than temporal inconsistencies in the Celeb-DF-V2 dataset. Audio detection exhibited the most stable statistical profile ($CV = 0.0151$), with no significant

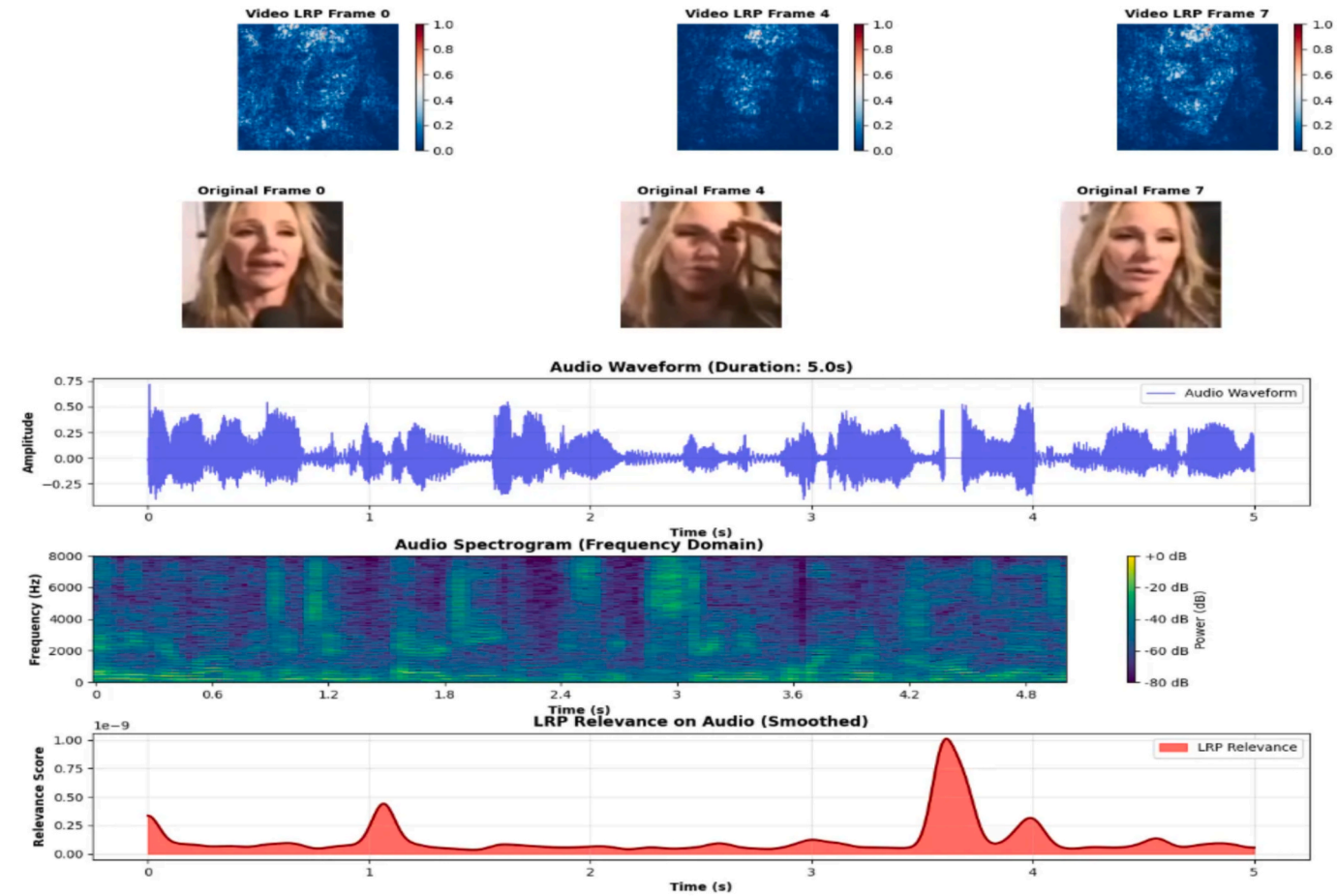


Fig. 35. Gradio – Multimodal Deepfake LRP.

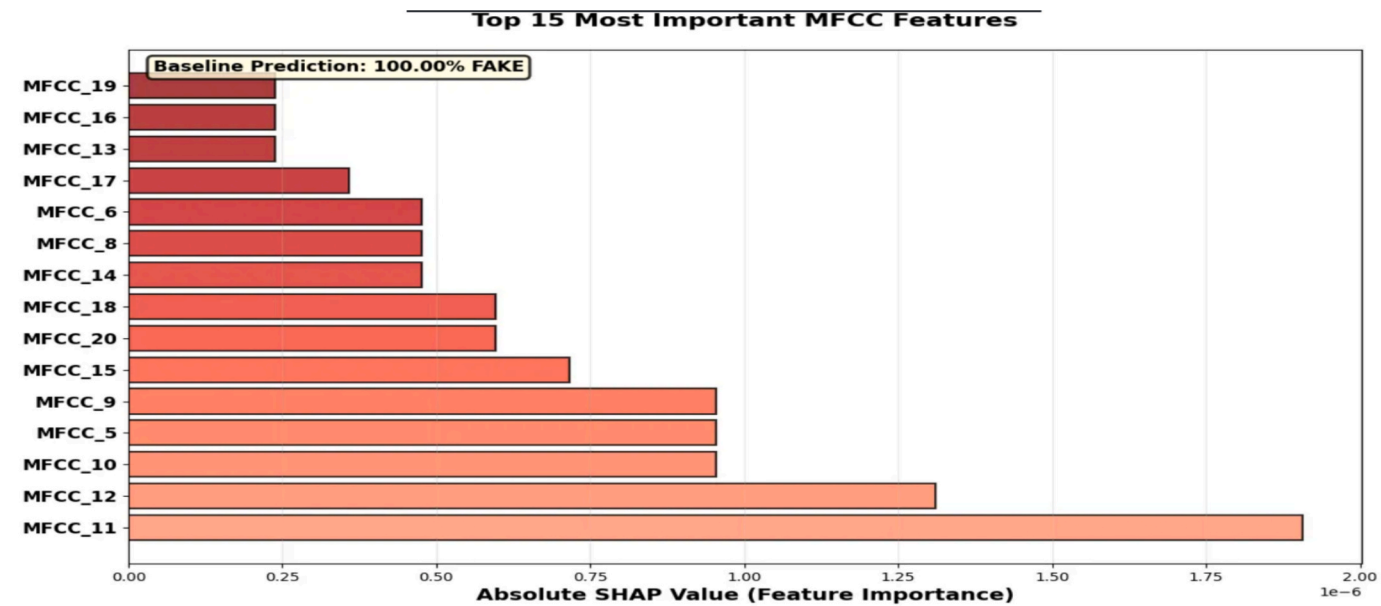


Fig. 36. Gradio – Multimodal Deepfake SHAP.

difference observed between XGBoost and CNN-BiLSTM ($p = 0.5000$), highlighting that both approaches provide complementary perspectives on audio forgery.

Ablation studies confirm that cross-attention fusion yields significant

gains compared to simple concatenation, particularly in precision (+3.21%) and specificity (+3.24%). The Friedman test ($p = 0.0035$) further validates that multimodal fusion captures additional information beyond individual modalities. Positive insertion–deletion AUC

Table 12
Comparison with deepfake detection methods.

Name	Model Name	XAI	Accuracy (%)	Dataset
[14]	DenseNet121	GradCAM	93.42	Image
[15]	CNN + MHSA	Attention	97.98	Image
[16]	VGG, Xception, RegularizedConvNet, RegularizedConvDenseNet	LIME	94.5	Image
[17]	InceptionResNetV2	LIME	99.87	Image
[18]	MobileNet	—	98.5	Image
Ours	InceptionV3	Saliency, LIME	99	Image
[10]	Ensemble	Grad-CAM	92.20	Video
[19]	EfficientNet	LIME, RISE, SOBOL, GradCAM, SHAP	96.26	Video
[20]	Ensemble	—	96.33	Video
[21]	CNN + CapsuleNet + LSTM	Grad-CAM	88.0	Video
[22]	Xception	LRP, LIME	94.33	Video
Ours	DenseNet169(Frame) Hybrid Model (CNN + BiLSTM)	GradCAM	96.76	Video
[23]	ML	SHAP	94.67	Audio
[24]	VGG16	SHAP, LIME, Grad-Cam	93.37	Audio
[25]	CNN – LSTM-perm	LRP, Deep Taylor, Integrated Gradients	98.97	Audio
[26]	AST, GBDT, Wav2Vec	Occlusion, Attention Visualization	F1-Score = 85	Audio
[27]	Fusion	SHAP, LRP	eer = 3.39	Audio
Ours	XGBoost, Hybrid Model (CNN + BiLSTM)	SHAP, LIME, LRP, GradCAM	98.87	Audio
[28]	BA-TFD, BA-TFD+	—	96	Multi-modal
[29]	CNN + ViT	—	98.12	Multi-modal
[30]	X3A	—	99.60	Multi-modal
[31]	CLFormer	—	99.33	Multi-modal
[32]	EfficientNetB0 + MFCC + BiLSTM	GradCAM, SHAP	87.25	Multi-modal
[33]	DiMoDif	—	99	Multi-modal
[34]	ERF-BA-TFD+	—	96.30	Multi-modal
Ours	Hybrid Model (CNN + BiLSTM + Cross Attention)	SHAP, LRP, GradCAM Integrated Gradients	96.32	Multi-modal

differences across all modalities demonstrate that the regions highlighted by XAI truly influence model decisions.

From a deployment perspective, the framework can operate on consumer-level hardware (GTX 1050, 8 GB RAM). Compared with higher-accuracy multimodal models such as X3A and CLFormer, the primary distinction of the proposed framework lies in its systematic explainability design, providing both visual and textual outputs across all modalities. This capability makes it particularly suitable for accountability-critical applications such as digital forensics and public policy.

Cross-dataset evaluation against FakeAVCeleb v1.2 further contextualizes the generalization behavior of the multimodal model. The Wilcoxon-confirmed dominance of the video channel (87.00% vs. audio 71.00%) in the statistical analysis aligns with the cross-dataset finding that audio-driven discriminative signals do not transfer effectively to datasets where audio manipulation is absent or stylistically different. This modality asymmetry suggests that future architectures should

incorporate domain-invariant feature extraction, particularly for the audio branch, to improve robustness across heterogeneous forgery types.

5. Conclusion

In this study, a unified and explainable deepfake detection framework has been proposed to address the growing challenges posed by increasingly realistic and multimodal deepfake content. Unlike conventional approaches that focus on a single modality, the proposed system enables comprehensive analysis across image, video, audio, and audio-visual data within a single framework. By integrating modality-specific deep learning models with a cross-attention-based multimodal fusion mechanism, the study achieves high detection performance while maintaining consistency across different data types. In addition, a modality-adaptive XAI strategy has been developed, combining SHAP, LIME, LRP, and Grad-CAM to provide both visual and textual explanations. This unified explainability design enhances transparency and supports practical usability through an interactive interface.

The main contribution of this work lies in jointly addressing three critical challenges in the literature: multimodal deepfake detection, effective cross-modal feature fusion, and unified explainability. The proposed framework not only improves detection accuracy across multiple benchmark datasets but also provides interpretable outputs, making it suitable for real-world applications such as digital forensics and media verification. Despite these contributions, several limitations should be acknowledged. First, the proposed framework relies on supervised learning and its performance may be affected by the diversity and quality of the training datasets. Second, although the multimodal model captures audio-visual relationships, it is limited to two modalities and does not incorporate other potential data sources such as text or metadata. Third, the computational cost of maintaining multiple modality-specific models may limit scalability in resource-constrained environments. Finally, while the explainability methods provide meaningful insights, they remain post-hoc and may not fully reflect the internal reasoning of deep learning models.

Future research can extend this work in several directions. Developing fully integrated end-to-end multimodal architectures that jointly learn from all modalities could further improve performance and efficiency. Incorporating additional modalities such as textual data or contextual metadata may enhance robustness against sophisticated deepfake attacks. Moreover, exploring inherently interpretable model designs and real-time deployment strategies would increase the applicability of such systems in real-world scenarios. Finally, evaluating the proposed framework on more diverse and large-scale datasets would further validate its generalizability. In conclusion, this study contributes to the advancement of deepfake detection by providing a comprehensive, interpretable, and multimodal framework, addressing key limitations in existing approaches and offering a foundation for future research in this rapidly evolving field.

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CRedit authorship contribution statement

Senanur Armağan: Writing – original draft, Software, Data curation, Conceptualization. **Esra Gündoğan:** Methodology, Investigation. **Mehmet Kaya:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation. **Reda Alhajj:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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